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Cluster size determines internal structure of transcription factories in human cells

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eLife Assessment

This is a **valuable** polymer model that provides insight into the origin of macromolecular mixed and demixed states within transcription clusters. The simulations are well performed and clearly presented in the context of existing experimental datasets. This **compelling** study will be of interest to those studying gene expression in the context of chromatin.

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Abstract

Transcription is a fundamental cellular process, and the first step of gene expression. In human cells, it depends on the binding to chromatin of various proteins, including RNA polymerases and numerous transcription factors (TFs). Observations indicate that these proteins tend to form macromolecular clusters, known as *transcription factories*, whose morphology and composition is still debated. While some microscopy experiments have revealed the presence of *specialised factories*, composed of similar TFs transcribing families of related genes, sequencing experiments suggest instead that mixed clusters may be prevalent, as a panoply of different TFs binds promiscuously the same chromatin region. The mechanisms underlying the formation of specialised or mixed factories remain elusive. With the aim of finding such mechanisms, here we develop a chromatin polymer model mimicking the chromatin binding-unbinding dynamics of different types of complexes of TFs. Surprisingly, both specialised (i.e., demixed) and mixed clusters spontaneously emerge, and which of the two types forms depends mainly on cluster size. The mechanism promoting mixing is the presence of non-specific interactions between chromatin and proteins, which become increasingly important as clusters become larger. This result, that we observe both in simple polymer models and more realistic ones for human chromosomes, reconciles the apparently contrasting experimental results obtained. Additionally, we show how the introduction of different types of TFs strongly affects the emergence of transcriptional networks, providing a pathway to investigate transcriptional changes following gene editing or naturally occurring mutations.

Introduction

The 3D organization of chromatin, the filament composed of DNA wrapped around histone proteins which constitutes the building block of mammalian chromosomes, is a dynamic and intricate blueprint that is thought to be important for cellular function and gene expression [1]. Recent advances in microscopy and highthroughput sequencing [2] have revealed a rich hierarchy of 3D chromatin structures within the cell nucleus. These range from relatively small DNA loops of tens to hundreds of base pairs (bps), to large organised domains spanning over hundreds of

thousands of base pairs (or kilo-base pairs, kbp), which are referred to as topologically-associating domains (TADs), whose segments interact more frequently among each other than with other parts of the genome. At even larger scales, the genomic material divides into A (active) and B (inactive) compartments, which have different gene activity and 3D compaction, whereas different chromosomes occupy distinct territories inside the nucleus [3–5].

A central question in cellular biology is the extent to which this rich and multi-scale organization is influenced, or even driven, by transcription [6], the fundamental biological process during which the information encoded in a segment of DNA is converted into RNA, to be then translated into proteins. On the one hand, it is widely believed that TADs remain largely invariant in cells with very different transcriptional programs (e.g., belonging to different organs) – which points to little role for transcription in determining structure [7] (for an opposing view, see [8]). On the other hand, enzymes engaged in the process of transcription, known as RNA polymerases, tend to form aggregates inside the nucleus, often referred to as phase-separated condensates, hubs, or transcription factories [6, 9–12]. Being attached to a factory strongly enhances the transcriptional activity of a gene [6, 9], therefore factories are a primary example of a structural unit with a clear transcriptional role.

A recent effective way to investigate this intricate connection between transcription and 3D chromatin structure has been provided by polymer models together with Brownian dynamics simulations [13–25]. This *in silico* approach has pointed to a simple and generic mechanism – the bridging-induced attraction or bridging induced phase separation – that spontaneously drives formation of transcription factories [26, 27]. Such microphase separation is due to the fact that, in this type of modelling, TF and polymerase complexes (TF:pol) are usually depicted as multivalent elements, so that each of them can simultaneously bind to several chromatin sites: this is reasonable as a complex of proteins can easily have more than one chromatin-binding domain. Multivalent binding triggers a positive feedback: a TF:pol binding to the chromatin filament provokes a local increase of chromatin density as it attracts several chromatin sites. The higher chromatin density, in turn, recruits further TF:pol resulting in a cluster, or transcription factory, formation. This feedback is eventually arrested by entropic costs associated with crowding and looping more and more DNA [27, 28].

The existence of clusters prompts the question: does a typical cluster mainly contain just one kind of TF, or many different ones? On the one hand, microscopy experiments suggest that different factories specialize in transcribing different sets of genes, so that any one factory typically contains mainly one type of TF. For example, active forms of RNA polymerases II and III are each housed in distinct nucleoplasmic factories that make genic and snRNA transcripts respectively [9, 29, 30]. Similarly, distinct ERα, KLF1, and NFκB factories specialize in transcribing genes involved in the estrogen response, globin production, and inflammation [31–33]. One important consequence of the formation of such specialized factories is the creation of 3D networks [34], in which genes sharing the same TFs are co-transcribed in the same clusters.

On the other hand, and in contrast to the evidence for specialized factories, chromatin immunoprecipitation (ChIP) techniques have revealed the existence of particular chromatin fragments, called “highly-occupied targets” (or HOT), which are promiscuously bound by several different TFs [35–37]. Additionally, single-cell transcriptional profiling points to expression levels varying continuously as cells differentiate into other cell types, which points to a complex interplay between many factors, rather than a few acting as binary switches [38, 39]. Interestingly, simulations of the types described above which involve 2 different kinds of TFs, each one binding specifically to two different TU types, show the resulting clusters often contain bound TFs of just one type, rather than mixtures, although this aspect has not been investigated in depth [11, 13, 21, 23, 26, 27, 40–48].

Here, we develop a polymer model with the aim of investigating the mechanisms leading to the formation of specialised or mixed factories: i.e., clusters formed by a single type or by multiple types of TFs respectively. Within our framework, chromatin is depicted as a polymer composed of a multicolour sequence of beads, corresponding to transcription units (TUs) of different types (or colours), each one binding to the corresponding type of transcription factors, represented as

additional spheres diffusing in the system. With respect to previous works on multicolour models [22, 27, 49–52], there are two important differences. First, here we model chromatin transcription by making the assumption that a chromatin bead is transcribed when it binds to a TF. In this way we are able to investigate the link between 3D structure of active chromatin and transcription (transcriptional patterns and emerging transcriptional correlation networks), rather than solely structure as done in previous models. Second, we study the morphology of the ensuing transcriptional clusters, studying their composition and the way in which it can be affected by the 1D binding landscape, and the balance between non-specific and specific chromatin-TF interactions.

Our main result is that specialised (demixed) and mixed clusters are *not* mutually exclusive. More specifically, we unexpectedly find a transition, or crossover, between a specialised and a mixed clusters regime, influenced by the size of emerging clusters. Smaller clusters are typically specialised, whereas larger clusters are more likely to be formed by different TF types. This result enables us to reconcile the apparently contrasting experimental observations cited above: it is no longer surprising that specialised and mixed clusters can coexist within the same cell, as cluster size depends, for instance, on the local 1D pattern of TU binding sites along chromatin. It is also important to specify that, even if an investigation of a transition between small and larger transcription factories have been carried out in mouse embryonic stem cells [53], in serum-stimulated cells [54] and in zebrafish embryos [55], our study aims to provide a broader mechanistic overview. We further integrated our multi-color model with experimental data, specifically DNase hypersensitive site (DHS) locations, to study human chromosomes. Here, two colours are considered as the simplest extension of the previous DHS model with one color [11], by distinguishing between active TF:pol complex that bind respectively to cell-type-invariant and cell-type-specific TUs in strings mimicking whole human chromosomes. Finally, the existence of specialized and mixed factories is further validated by incorporating different types of proteins into more complex and realistic chromatin models, such as the “highly predictive heteromorphic polymer model” (HiP-HoP model), which accounts for loop extrusion by cohesin-like complexes, the presence of inactive or silenced chromatin, and chromatin heteromorphism [41].

Results

Toy model with different transcription factors

To try to solve the apparently contrasting views of segregated and mixed factories we start by introducing a new simple polymer model, where a 3 Mbp chromatin fragment is represented by a chain of 1000 beads (each 30 nm in diameter, and corresponding to 3 kbp). Different kinds of TU beads are positioned regularly in this string, but are coloured randomly yellow, red, or green (we refer to this case as the random string). Different kinds of TFs are modelled as diffusing spheres, at first approximation of the same size of chromatin beads (see later for simulations changing the TFs size), which bind reversibly and strongly to beads of the same colour, and weakly to all others (see Fig. 1A [↗](#), and Methods for further details). After running a Brownian-dynamics simulation, Fig. 1B [↗](#)

We now assume that the spheres represent TF:pol complexes, and make the reasonable assumption that a TU bead is transcribed if it lies within 2.25 diameters (2.25σ) of a complex of the same colour (small changes in this threshold do not affect qualitatively the measured activity, as shown in Figure S11); then, the transcriptional activity of each TU is given by the fraction of time that the TU and a TF:pol lie close together. Fig. 1D [↗](#) reports the mean activity profile down the string obtained combining transcriptional data from 100 independent simulation runs each lasting $8 \cdot 10^5 \tau_B$; TUs with the lowest activities are flanked by differently-coloured TUs, while those with the highest activities are flanked by similarly coloured TUs (dashed rectangles in Fig. 1D [↗](#)). As expected, a single-colour model with the same TU placement leads to a flat activity profile (Figure S1A). Clearly, close proximity in 1D genomic space favours formation of similarly coloured clusters.

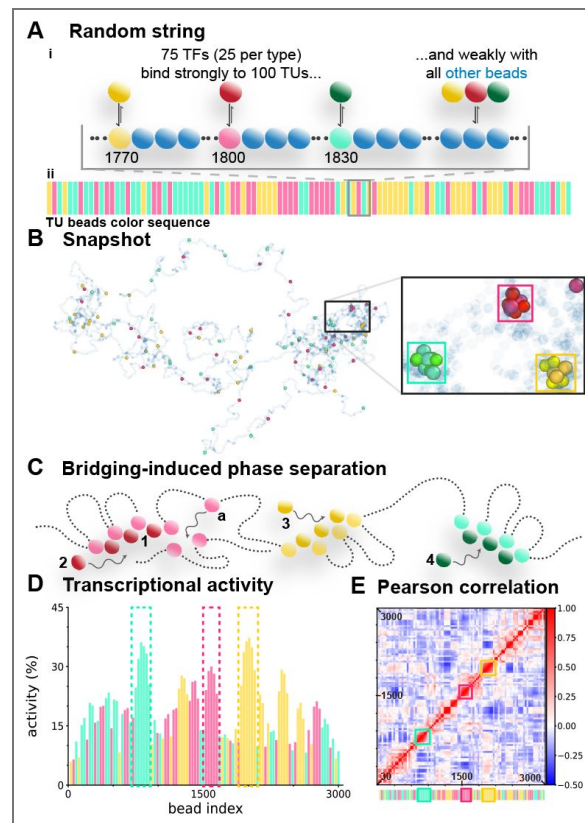


Figure 1. Toy model, with TUs coloured randomly (the random string).

(A) Overview. (i) Yellow, red, and green TFs (25 of each colour) bind strongly (when in an on state) to 100 TUs beads of the same colour in a string of 3000 beads (representing 3 Mb), and weakly to blue beads. TU beads are positioned regularly and coloured randomly, as indicated in one region of the string. TFs switch between off and on states at rates $\alpha_{off} = 10^{-5} \tau_B^{-1}$ and $\alpha_{on} = \alpha_{off}/4$ (τ_B Brownian time, which one can map to 0.6×10^{-3} s, see SI). (ii) The sequence of bars reflects the random sequence of yellow, red, and green TUs (blue beads not shown). **(B)** Snapshot of a typical conformation obtained after a simulation (TFs not shown). Inset: enlargement of boxed area. TU beads of the same colour tend to cluster and organize blue beads into loops. **(C)** Bridging-induced phase separation drives clustering and looping. Local concentrations of red, yellow, and green TUs and TFs might appear early during the simulation (blue beads not shown). Red TF 1 – which is multivalent – has bound to two red TUs and so forms a molecular bridge that stabilizes a loop; when it dissociates it is likely to re-bind to one of the nearby red TUs. As red TU 2 diffuses through the local concentration, it is also likely to be caught. Consequently, positive feedback drives growth of the red cluster (until limited by molecular crowding). Similarly, the yellow and green clusters grow as yellow TF 3 and green TF 4 are captured. **(D)** Bar heights give transcriptional activities of each TU in the string (average of 100 runs each lasting $8 \times 10^5 \tau_B$). A TU bead is considered to be active whilst within $2.24\sigma \sim 6.7 \times 10^{-8} m$ of a TF:pol complex of similar colour. Dashed boxes: regions giving the 3 clusters in the inset in **(B)**. **(E)** Pearson correlation matrix for the activity of all TUs in the string. TU bead number (from low to high) is reported on axes, with pixel colour giving the Pearson value for each bead pair (bar on right). Bottom: reproduction of pattern shown in **(A,ii)**. Boxes: regions giving the 3 clusters in the inset in **(B)**. shows a typical 3D conformation found in the steady state. Remarkably, clusters of TUs and TFs with distinct colours appear and disappear spontaneously. Such clustering is driven by the positive feedback illustrated in Fig. 1C; it depends critically on TFs being able to form molecular bridges that anchor loops [11, 26, 27].

We next examine how closely transcriptional activities of different TUs correlate [56]; the Pearson correlation matrix for all TUs is shown in Fig. 1E. This is built in the following way: for every couple i - j of TUs, we evaluate the Pearson correlation coefficient between the transcription of i and j , and then we colour the ij (and ji) pixel of the matrix accordingly. Correlations between neighbouring TUs of similar colour are often positive and strong, resulting in square red blocks along the diagonal (coloured boxes in Fig. 1E highlight the 3 clusters shown in the zoom in Fig. 1B). This effect is again due to the self-assembly of clusters containing neighbouring TUs of the same colour. In contrast, neighbours with different colours tend to compete with each other for TF:pol, and so down-regulate each other to yield smaller correlations. Correlations are more trivial in the single-color counterpart of Fig. 1, where the matrix yields only a positive-correlation band along the diagonal (Figure S1B). These results provide simple explanations of two mysterious effects – the first being why adjacent TUs throughout large domains tend to be cotranscribed so frequently [57]. The second concerns how expression quantitative trait loci (eQTLs)—genomic regions that are statistically associated with variation in gene expression levels—function. While current models often attribute their effects to post-transcriptional regulation through complex mechanisms [11, 58], we have previously argued that any transcriptional unit (TU) can act as an eQTL by directly influencing gene expression at the transcriptional level [11]. Here, we observe individual TUs up-regulating or down-regulating the activity of others TUs – hallmark behaviors of eQTLs that can transcriptional clusters, studying their composition and the way in which it can be affected by the 1D binding landscape, and the balance between non-specific and specific chromatin-TF interactions.

Our main result is that specialised (demixed) and mixed clusters are *not* mutually exclusive. More specifically, we unexpectedly find a transition, or crossover, between a specialised and a mixed clusters regime, influenced by the size of emerging clusters. Smaller clusters are typically specialised, whereas larger clusters are more likely to be formed by different TF types. This result enables us to reconcile the apparently contrasting experimental observations cited above: it is no longer surprising that specialised and mixed clusters can coexist within the same cell, as cluster size depends, for instance, on the local 1D pattern of TU binding sites along chromatin. It is also important to specify that, even if an investigation of a transition between small and larger transcription factories have been carried out in mouse embryonic stem cells [53], in serum-stimulated cells [54] and in zebrafish embryos [55], our study aims to provide a broader mechanistic overview. We further integrated our multi-color model with experimental data, specifically DNase hypersensitive site (DHS) locations, to study human chromosomes. Here, two colours are considered as the simplest extension of the previous DHS model with one color [11], by distinguishing between active TF:pol complex that bind respectively to cell-type-invariant and cell-type-specific TUs in strings mimicking whole human chromosomes. Finally, the existence of specialized and mixed factories is further validated by incorporating different types of proteins into more complex and realistic chromatin models, such as the “highly predictive heteromorphic polymer model” (HiP-HoP model), which accounts for loop extrusion by cohesin-like complexes, the presence of inactive or silenced chromatin, and chromatin heteromorphism [41].

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Local mutations

To explore the impact of introducing different colors, we characterize the effects of local mutations. We choose the most active region in the random string – one containing a succession of yellow TUs – and “mutate” 1–4 of these TUs by recolouring them red (Fig. 2A). These simulations are inspired by editing experiments performed using CRISPR/Cas9 [60]. Typical snapshots show red mutants are often ejected from yellow clusters (Fig. 2Bi), or cluster together to leave their wild-type neighbours in isolation (Fig. 2Bii). These changes are reflected in activity profiles (Fig. 2C; arrows indicate mutations). As the number of mutations in the cluster increase, activities of yellow beads in that cluster decrease (Fig. 2D), and new red clusters often emerge (Fig. 2B,ii; Fig. 2Ciii).

To confirm that 4 mutations in a yellow cluster often lead to the development of a red cluster, we monitor cluster dynamics over time. Fig. 2Ei illustrates a typical kymograph illustrating changes in activity of all TUs in the wild-type, corresponding to the same original string presented in Fig. 1; yellow, red, and green pixels mark activity of respective TUs, and black ones inactivity. In this particular simulation, a yellow cluster in the region that will be mutated (marked by the blue rectangle) is present during the first quarter of the time window; it then disappears to reappear half-way through the window and then persists until the end. From the activity profiles in Fig. 2C, we can observe that as the number of mutations increases, the yellow cluster is replaced by a red cluster, with the remaining yellow TUs in the region being expelled (Fig. 2Bii). This behavior is reflected in the dynamics, as seen by comparing panels Ei and Eii: in the string with four mutations, transcription of the yellow TUs is inhibited in the affected region, while prominent red stripes—corresponding to active, transcribing clusters—emerge (Fig. 2Eii). Pearson correlation matrices provide complementary information: the yellow cluster in the wild-type yields a solid red block indicating strong positive correlations (Fig. 2Fi), while most of the pixels of this block become light-red or white in the string with 4 mutations (Fig. 2Fii). These results confirm that local arrangements of TUs on the genetic map determine the extent to which any particular TU will cluster and so become active.

Variations in TF concentration

The concentration of TFs is expected to influence the global activity patterns observed and can be adjusted in our model accordingly. These simulations are motivated by experiments that reduce global TF levels using auxin-induced degrons [61]. Specifically, we reduce the concentration of yellow TFs binding to the random string by 30% (Fig. 3A). As expected, transcriptional activity falls both globally and locally (see yellow dotted rectangles in Fig. 3B and C). Surprisingly, activity of a nearby cluster of red TUs (numbers 1080, 1110, 1170, 1200, and 1530 to 1650) increases by 50% (red dotted rectangles in Fig. 3B and C). This effect is specific, in the sense that there is little effect on green clusters (e.g., compare Fig. 1D with Fig. 3B). We attribute this to a now-reduced steric competition for 3D space by yellow neighbours – fewer yellow clusters are present to stunt growth of nearby red ones. Fig. 3D shows the difference in correlation between the case with reduced yellow TFs and the case displayed in Fig. 1E. We can notice a change in correlation between the yellow cluster (boxed) and its neighbour clusters, even if distant. For instance, yellow clusters are more probable to be found both turned off due to the lack of yellow TFs, and thus their activation becomes more correlated. At the same time, when yellow clusters are turned off the activation of other clusters can be affected, with an increase or decrease of correlation. Overall, these results reveal numerous statistically significant correlations in gene activity both in proximal and distal regions of the genetic map. This observation aligns with the omnigenic model, which proposes that the manifestation of a genetic trait is influenced not only by the expression of core genes directly associated with the trait, but also by peripheral genes, which can exert indirect effects through gene regulatory networks. [58].

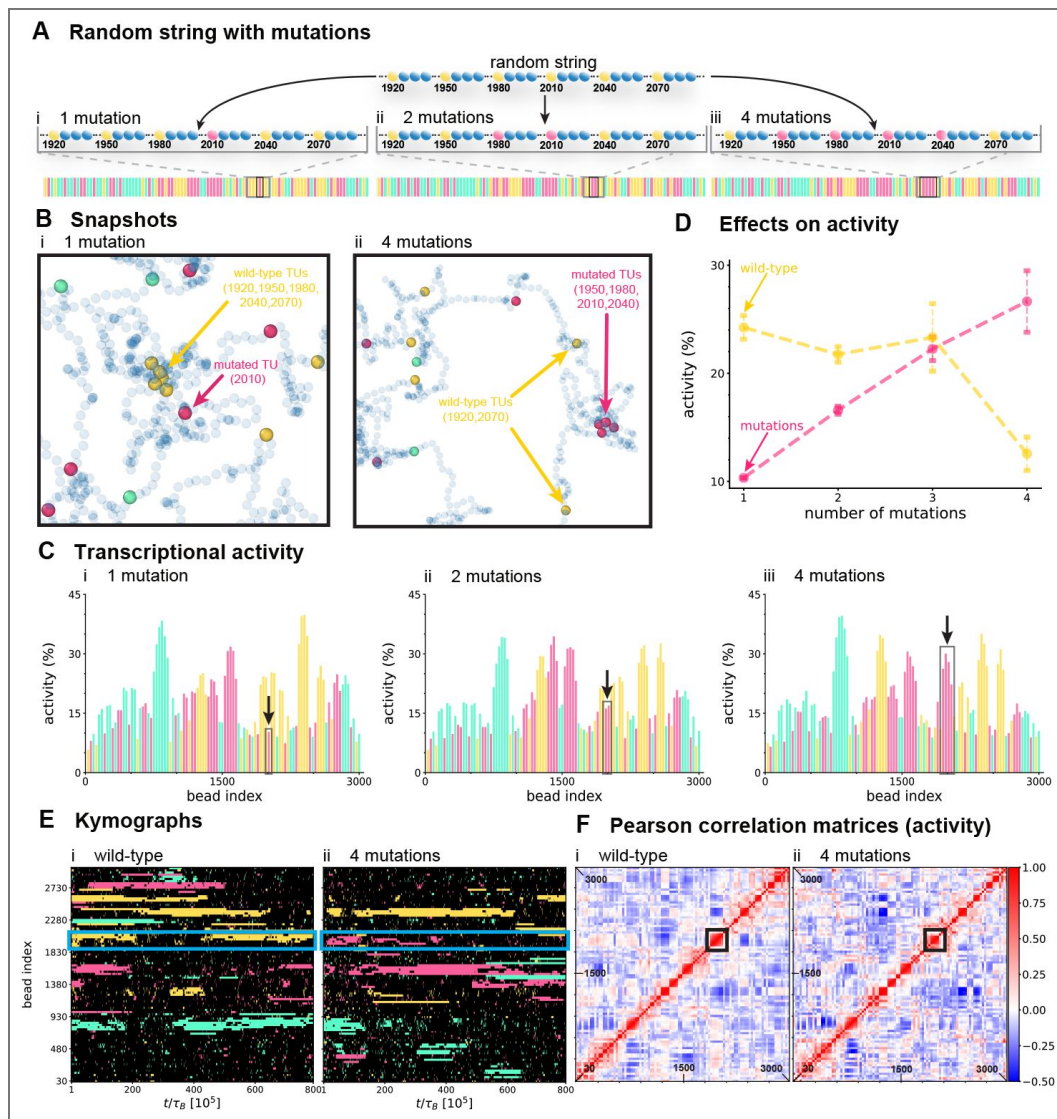


Figure 2. Simulating effects of mutations.

Yellow TU beads 1920, 1950, 1980, 2010, 2040 and 2070 in the random string have the highest transcriptional activity. 1-4 of these beads are now mutated by recolouring them red. **(A)** The sequence of bars reflects the sequence of yellow, red, and green TUs in random strings with 1, 2 and 4 mutations (blue beads not shown). Black boxes highlight mutant locations. **(B)** Typical snapshots of conformations with **(i)** one, and **(ii)** 4 mutations. **(C)** Transcriptional-activity profiles of mutants (averages over 100 runs, each lasting $8 \cdot 10^5 \tau_B$). Bars are coloured according to TU colour. Black boxes: activities of mutated TUs. **(D)** Activities ($\pm$ SDs) of wild-type (yellow) and different mutants. 3 mutations: TUs 1950, 1980 and 2010 mutated from yellow to red. **(E)** Typical kymographs for **(i)** wild-type, corresponding to the same original string presented in Figure 1, and **(ii)** 4-mutant cases, in which 4 yellow TUs have been mutated to red. Each row reports the transcriptional state of a TU during one simulation. Black pixels denote inactivity, and others activity; pixels colour reflects TU colour. Blue boxes: region containing mutations. **(F)** Pearson correlation matrices for wild-type and 4-mutant cases. Black boxes: regions containing mutations (mutations also change patterns far from boxes).

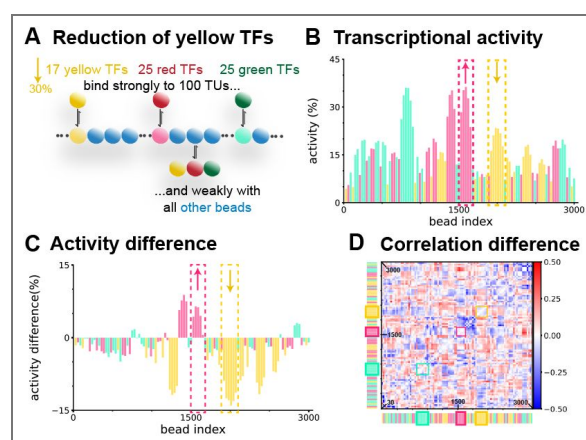


Figure 3. Reducing the concentration of yellow TFs reduces the transcriptional activity of most yellow TUs while enhancing the activities of some red TUs.

(A) Overview. Simulations are run using the random string with the concentration of yellow TFs reduced by 30%, and activities determined (means from 100 runs each lasting $8 \times 10^5 \tau_B$). (B) Activity profile. Dashed boxes: activities fall in the region containing the biggest cluster of yellow TUs seen with 100% TFs, as those of an adjacent red cluster increase. (C) Differences in activity induced by reducing the concentration of yellow TFs. This plot is obtained by subtracting the transcriptional activity of the wild-type, Figure 1D, from that of the current system in panel B. (D) Pearson correlation difference matrix. This plot is obtained by subtracting the Pearson correlation matrix of the wild-type, Figure 1E, from that of the current system. Boxes: regions giving the 3 clusters from Figure 1B, inset.

Effects of 1D TU patterns on transcriptional activity

To gain a deeper understanding of the local effects revealed by the random string, we now analyze and compare various toy strings that feature regular and repeating patterns of colored TUs (Fig. 4*A* and methods for further details). Two results are apparent. First, activities (Fig. 4*Bii*) in the 6-pattern case are higher overall (compare horizontal dotted lines), and more variable (compare activities of the two central TUs within each repeat with peripheral ones) relative to the 1-pattern case (Fig. 4*Bi*). This is consistent with positive additive effects acting centrally within each 6-pattern repeat, coupled to competitive negative effects of flanking and differently-coloured repeats at the edges. Second, the 6-pattern also has a Pearson correlation matrix (Fig. 4*Cii*) that is highly-structured, with a checkerboard pattern; red blocks on the diagonal indicate high positive correlations (so the 1D 6-pattern clearly favours 3D clustering). [Such a checkerboard pattern is not seen with a singlecolor model that has a correlation matrix with one red continuous diagonal when TUs are regularly spaced (Figure S1).] Additionally, blue off-diagonal blocks indicate repeating negative correlations that reflect the period of the 6-pattern. These results show how strongly TU position in 1D genomic space affect 3D clustering and activity, and that these effects depend on inclusion of more than one colour.

Emergent transcriptional correlation networks

We have seen many positive and negative correlations between activities of TUs in the random string (Fig. 1*C*).

We now select significant correlations from Pearson correlation matrices (those which are > 0.2 , Fig. 5*A*) to highlight emergent interaction networks [11]. In such networks, nodes represent each TU from first to last (other beads are not shown), and edges indicate positive (black) or negative (grey) correlations in activities of node pairs. Even for the toy random string, these networks prove to be very complex (Figure S2A). They are also “smallworld” (i.e., most nodes can be reached from other ones by a few steps [11, 62]). Given this complexity, we now consider simplified versions. Thus, in Fig. 5*Ai*, only interactions between red TUs are shown (the first red TU is at position 60, the last at position 2910, and interactions between different colours are not depicted). As expected, activities of most red TUs are positively correlated with those of nearby TUs. Conversely, negative correlations connect distant TUs, as found in the singlecolor model [11]; as we have seen, binding of red TFs to any red cluster reduces the number available to bind elsewhere.

In Fig. 5*Aii*, we consider just interactions between red TUs and green TUs. Remarkably, close-range positive correlations (black edges) are still seen between TU pairs that no longer bind TUs of the same colour. We suggest this is due to the presence of weakly-binding beads. Specifically, a red cluster organises a surrounding cloud of weakly-binding beads, and these will bind some green TFs that – in turn – bind green TUs. In contrast to the same-colour network in Figure 5*Ai*, there are now more long-range positive correlations, showing that the presence of multiple colors enriches the emerging network.

To obtain further quantitative insight into these subtle yet remarkable correlations, we compute the average of those between same- and different-colour TUs as a function of genomic separation (Fig. 5*B*). For the random string, same-colour correlations switch from clearly positive to slightly negative at about 300 beads (Fig. 5*Bi*, red curve). Differently-coloured correlations yield a broadly similar switch, although positive and negative values are weaker (Fig. 5*Bi*, blue curve). The 6-pattern gives qualitatively similar trends, with the magnitude of differently coloured correlations dampened further (Fig. 5*Bii*). In contrast, the 1-pattern string yields largely overlapping curves (Fig. 5*Biii*). These results illustrate how the sequence of TUs on a string can strikingly affect formation of mixed clusters; they also provide an explanation of why activities of human TUs within genomic regions of hundreds of kbp are positively correlated [63].

To quantify the extent to which TFs of different colours share clusters, we introduce a demixing coefficient, θ_{dem} (see Methods for definition), which can vary between 0 and 1. If $\theta_{\text{dem}} = 1$, a cluster contains only TFs of one colour (and so is fully demixed); if $\theta_{\text{dem}} = 0$, it contains both red

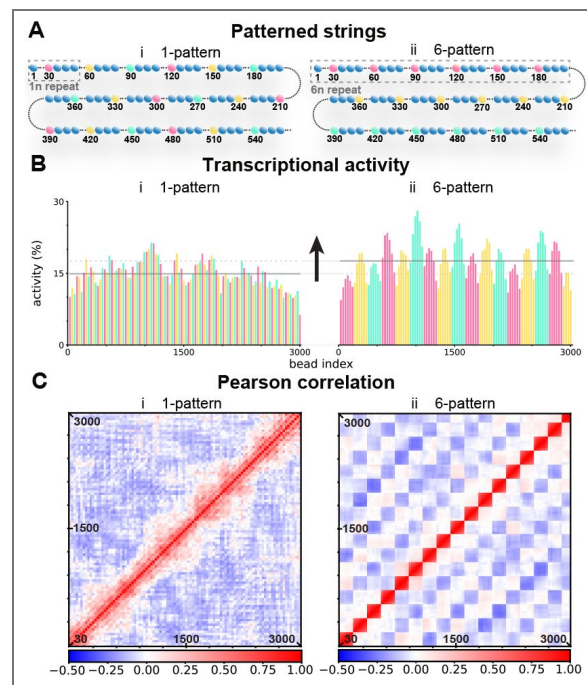


Figure 4. Clustering similar TUs in 1D genomic space increases transcriptional activity.

(A) Simulations involve toy strings with patterns (dashed boxes) repeated 1 or 6 times. Activity profiles plus Pearson correlation matrices are determined (100 runs, each lasting $8 \cdot 10^5 \tau_B$). (B) The 6-pattern yields a higher mean transcriptional activity (arrow highlights difference between the two means). (C) The 6-pattern yields higher positive correlations between TUs within each pattern, and higher negative correlations between each repeat.

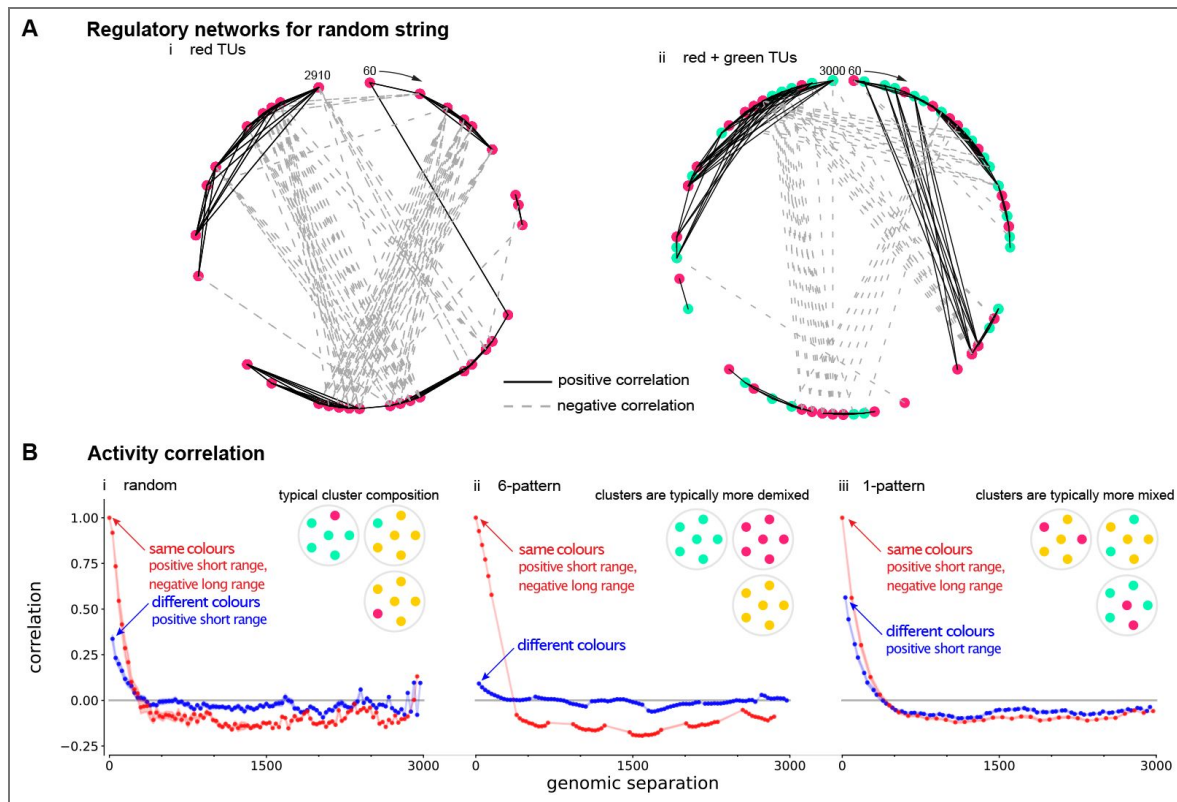


Figure 5. TU transcriptional networks and demixing.

Simulations are run using the toy models indicated, and complete correlation networks (qualitatively reminiscent of gene regulatory networks) constructed from Pearson correlation matrices. respectively (above a threshold of 0.2, corresponding to a p -value $\sim 5 \cdot 10^{-2}$). The complete network consists of $n = 100$ (A) Simplified network given by the random string. TUs from first (bead 30) to last (bead 3000) are shown as peripheral nodes (coloured according to TU); black and dashed grey edges denote statistically-significant positive and negative correlations, individual TUs, so that there are $n_c = \binom{100}{2} = 4950$ pairs of TUs couples; we find 990 black and 742 gray edges. Since p -value $n_c = 223$, most interactions (edges) are statistically significant. Networks shown here only correlations (i) between red TUs, and (ii) between red and green TUs. (ii) (B) Average Pearson correlation (shading shows $\pm$ SD, and is usually less than line/spot thickness) as a function of genomic separation for the (i) random, (ii) 6-, and (iii) 1-pattern cases. Correlation values at fixed genomic distance are taken from super-/sub-diagonals of Pearson matrices. Red dots give mean correlation between TUs of the same color (3 possible combinations), and blue dots those between TUs of different colors (4 possible combinations). Cartoons depict contents of typical clusters to give a pictorial representation of mixing degree (as this determines correlation patterns); see SI for exact values of θ_{dem} .

and green TFs in equal numbers (and so is fully mixed). Intuitively, one might expect θ_{dem} to fall as the number of adjacent TUs of similar colour in a string fall; this is what is seen with the 6- and 1-patterns – strings with the most and least numbers of adjacent TUs of similar colour, respectively (Figure S2B; shown schematically by the cluster cartoons in Fig. 5B).

Our simulations then show that in cases where same- and different-colour Pearson correlations trends overlap (as in the 1-pattern string, see Fig. 5Biii), clusters are more mixed (have a larger value of θ_{dem}). Instead, in cases where same- and different-colour correlations trends do not overlap, or are more different (as in the 6-pattern string, see Fig. 5Bii), then clusters are typically unmixed, and so have a larger value of θ_{dem} (Figure S2B). These results on activity correlation and TF cluster composition suggest that, if eQTLs act transcriptionally as expected [11], down-regulating eQTLs are likely to be located further from their target genes than up-regulating ones. In addition, it is important to note that mixing is promoted by the presence of weakly binding beads; replacing these with non-interacting ones leads to increased demixing and a reduction in long-range negative correlations (Figure S3). More generally, our findings indicate that the presence of multiple TF colors offers an effective mechanism to enrich and fine-tune transcriptional regulation.

Transcriptional activity and comparison with real human chromosomes

We now simulate human chromosome 14 (HSA 14) in HUVECs, with individual beads in the string coloured appropriately (Fig. 6A). Thus, TUs transcribed uniquely in HUVECs are coloured red, housekeeping TUs (i.e., ones also expressed in a stem cell, namely H1-hESCs) are green, euchromatic regions blue, and heterochromatic ones grey. Fig. 6B shows a typical snapshot; red and green clusters again form spontaneously. We next determine transcriptional activities, rank them in order from high to low, and compare binned rank orders with those obtained experimentally by GRO-seq (Fig. 6C); most counts lie along the diagonal, meaning there is a good agreement between the two data sets. More quantitatively, Spearman's rank correlation coefficient is $3.66 \cdot 10^{-1}$, which compares with $3.24 \cdot 10^{-1}$ obtained previously using a single-colour model [11]. In both cases the estimated uncertainty is of order 10^{-3} (mean and SD obtained using the bootstrap technique over 100 trials) and the p -value is $< 10^{-6}$ (2-sided t-test), showing statistical significance.

Activity predictions are also improved compared to the one-colour model with HSA 18 and HSA 19 in HU-VECs, plus HSA 14 in GM12878 (Figure 6D and Figure S4). However, Spearman's rank coefficient for gene-poor HSA 18 is about twice that for gene-rich HSA 19; this may be due to additional regulatory layers in regions with high promoter density. These results show that our multicolour polymer model generates strings that can mimic structures and functions found in whole chromosomes. Additionally, simulated contact maps show a fair agreement with Hi-C data (Figure S5), with a Pearson correlation $r \sim 0.7$ (p -value $< 10^{-6}$, 2-sided t-test). However, because this 2-color model does not include heterochromatin-binding proteins and cohesin-mediated active loop-extrusion, as in the Hip-Hop model later discussed, we should not expect a very accurate reproduction of Hi-C maps.

Specialized and mixed clusters

Inspection of simulation snapshots shows 1-colour clusters tend to be smaller than mixed (2-colour) ones (Fig. 7A). To quantify this, we count numbers and types of TFs in individual clusters (Figures 7B and S7). Clusters with just two bound TFs never contain both colours; conversely, those with > 20 bound TFs never contain just one colour (Fig. 7B). We also measure the average value of the demixing coefficient, θ_{dem} (Materials and Methods). If $\theta_{\text{dem}} = 1$ ($\theta_{\text{dem}} = 0$), this means that a cluster contains only TFs of one colour (a mixture of TFs) and so is fully demixed (maximally mixed). The result is shown in Fig. 7C, and shows the emergence of a crossover between a demixed regime, corresponding to single-colour clusters, and a mixed regime, corresponding to multiple-colour clusters, which is triggered by an increase in cluster size. [Note that we speak of a crossover between regimes, rather than a phase transition, as clusters are

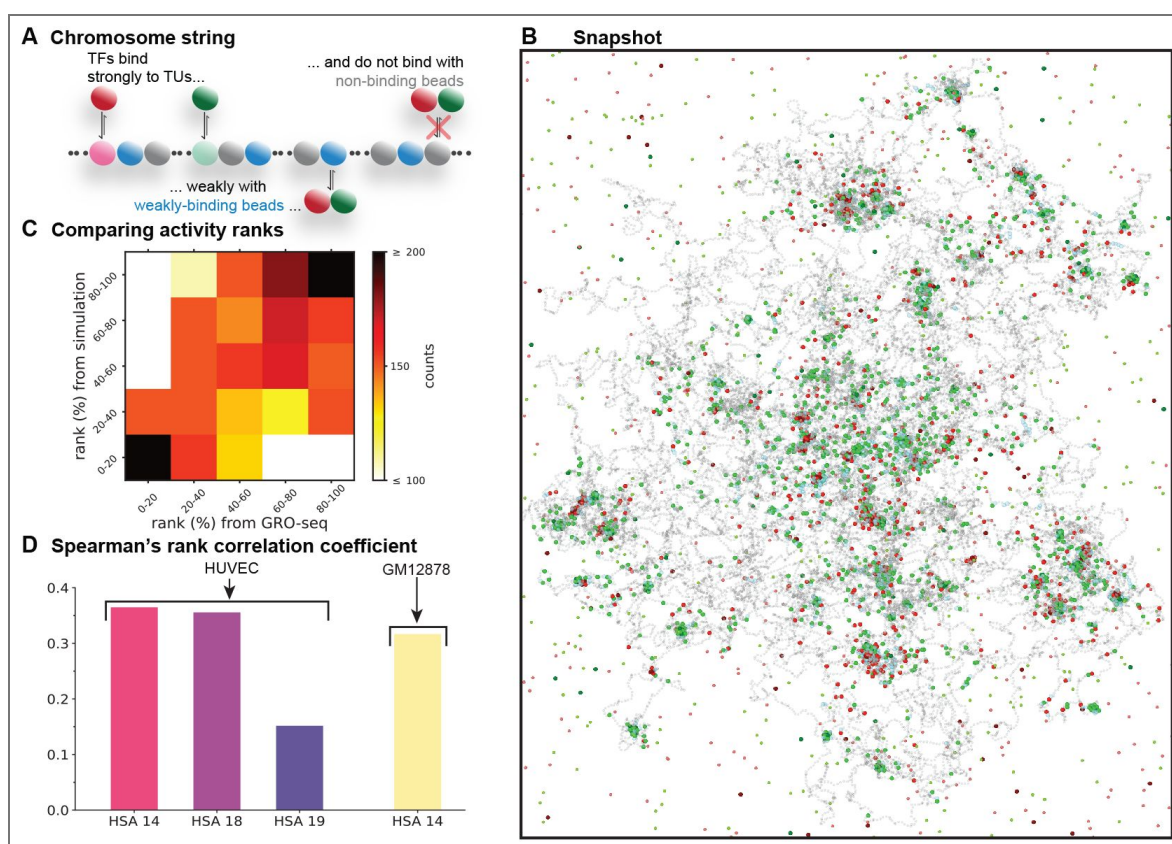


Figure 6. Comparison of transcriptional activities of TUs on different human chromosomes determined from simulations and GRO-seq.

(A) Overview of panels (A-C). The 35784 beads on a string representing HSA14 in HUVECs are of 4 types: TUs active only in HUVECs (red), “house-keeping” TUs – ones active in both HUVECs and ESCs (green), “euchromatic” ones (blue), and “heterochromatic” ones (grey). Red and green TFs bind to TUs of the same colour with strong (specific) interactions, while they experience a weak (non-specific) attraction to euchromatin. Interactions between both red and green TFs and heterochromatin are purely repulsive. (B) Snapshot of a typical conformation, showing both specialized and mixed clusters. (C) TU activities seen in simulations and GRO-seq are ranked from high to low, binned into quintiles, and activities compared. (D) Spearman's rank correlation coefficients for the comparison between activity data obtained from analogous simulations and GRO-seq for the chromosomes and cell types indicated.

finite-size and we do not consider any thermodynamic limit.] The cross-over point between fully mixed and demixed (where the average value of $\theta_{\text{dem}} = 0.5$) occurs when there are ~ 10 TFs per cluster (Fig. 7C): notably, this is similar to the average number of productively transcribing pols seen experimentally in a transcription factory [6]. Similar results are obtained for different cell types, or chromosomes (see Figs. S6 and S7 for the case of HSA 18, 19 in HUVEC, and HSA 14 in GM12878), and chromosomes under confinement (Fig. S10), with realistic chromatin densities. The latter situation suggests that, as far as the formation of transcription factories and the crossover between mixed and demixed clusters are concerned, chromatin density does not play a crucial role. Other phenomena can indeed depend on density, especially with respect to global chromatin structure (e.g., entanglements and rare long-range contacts). Additionally, simulations of HSA 14 in HUVEC cells with different size of TF:pol (0.5σ and 0.16σ) also lead to similar results (see Fig. S9). Importantly, the critical number of TFs per cluster separating demixed and mixed cluster is around ~ 10 in all these different cases. These results suggest that neither the sequence of TUs and its ratio to TFs (which varies among chromosomes, as for instance HSA 18 and HSA 19 are gene poor and gene rich respectively), nor the chromatin density affect the nature of the crossover between the regimes of demixed and mixed clusters.

The existence of a crossover between specialized (demixed) and mixed factories with increasing size is therefore a generic feature of our model, and it can be explained by the following physical argument depending on non-specific binding. Two red TFs in an unmixed cluster might stabilise 3 loops, and so bring into close proximity only a few non-specific binding sites that could bind a green TF. In contrast, 10 red TFs in a cluster will stabilise many loops that inevitably bring into close proximity many non-specific binding sites – and this makes it highly likely that some green TFs will also bind nearby to create a mixed cluster. The mixing crossover provides a way to reconcile observations that some clusters are unmixed (like factories rich in polymerases II and III), and others highly mixed (like HOTs). This is because clusters in a single cell are generally polydisperse, or differ in size (e.g., due to the local chromatin environment, or the patterning of TUs along the genome), hence mixed and specialised factories can coexist in the same nucleus. Note that cluster size is a key parameter because it strongly affects the balance between non-specific and specific chromatin-protein interaction.

Finally, as for the toy model, the balance between mixing and demixing determines correlation patterns. For example, activity patterns of same- and differently-colored TUs in the whole chromosome (Figure S8) are much like those in the 1-pattern model (Fig. 5Biii). We attribute this to $\sim 78\%$ TFs being in mixed clusters ($\theta_{\text{dem}} < 0.5$), and so inevitably the resulting interactions will dominate the pattern seen.

Our model is already inherently out of thermodynamic equilibrium, as it includes non-equilibrium switching between binding and non-binding states for chromatin-binding proteins, resembling ATP-dependent post-translational modification of such proteins[64]. There are though important principles of chromatin organisation which the presented model does not consider. First, an important remaining question is whether other active (ATP-consuming) processes naturally present in the nucleus, such as loop extrusion [65], affect the results we found. Second, following the same logic behind the multicolor polymer model presented here, it is interesting to ask whether the presence of additional types of inactive, as well as active, TFs and chromatin beads changes the picture.

To answer these questions, we have turned to a more complex framework, and to the HiP-HoP model, which includes loop extrusion by cohesin-like complexes and chromatin heteromorphism [41, 66], as well as accounting for inactive, as well as active, chromatin and TFs. Specifically, we performed simulations for HSA14 in HUVEC using a multicolor version of the HiP-HoP model (see SI for more details). A typical configuration is shown in Figure 8A, where grey regions represent locally compact regions (which are poor in H3K27ac), while cyan regions represent disrupted regions (which are enriched in decompacted chromatin and in H3K27ac). In addition, H3K27me3 and H3K9me3 data were used to determine the chromatin binding sites for polycomb-like and heterochromatin-associated proteins (such as HP1): these are represented in yellow and blue respectively. As in the previous DHS model, TUs only present in HU-VEC are represented in red,

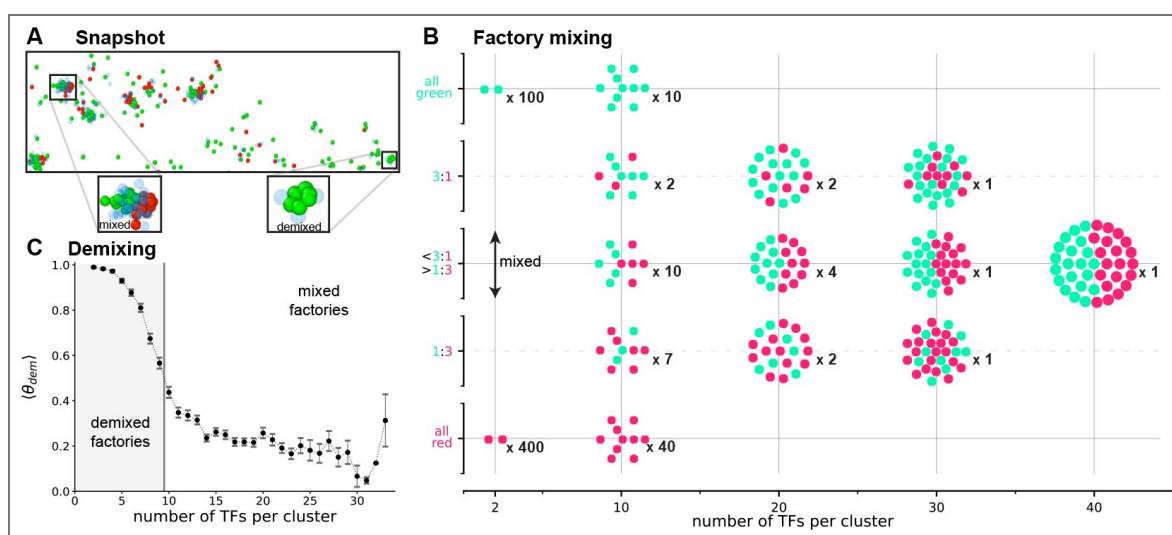


Figure 7. Small clusters tend to be unmixed, large ones mixed.

After running one simulation for HSA 14 in HUVECs, clusters are identified. **(A)** Snapshot of a typical final conformation (TUs, non-binding beads, and TFs in off state not shown). Insets: a large mixed cluster and a small demixed one. **(B)** Example clusters with different numbers of TFs/cluster (2, 10, 20, 30, 40) chosen to represent the range seen from all-red to all-green (with 3 intervening bins). Black numbers: observed number of clusters of that type seen in the simulation. **(C)** Average of the demixing coefficient θ_{dem} (error bars: SD), showing a crossover between demixed and mixed clusters with increasing cluster size. Values of 1 and 0 are completely demixed and completely mixed respectively. Grey area: demixed regime where $\theta_{dem} > 0.5$.

while the house-keeping ones in green. Inspection of simulation snapshots shows the presence of small clusters that are demixed (Fig 8C) and large cluster that are mixed (Fig. 8B). We also measure the average value of the demixing coefficient, θ_{dem} (Fig. 8 D). As in the simpler DHS model, the crossover point between fully mixed and demixed (where the average value of $\theta_{\text{dem}} = 0.5$) occurs when there are ~ 10 TFs per cluster. These simulations further confirm the robustness and generality of our results regarding the mixing-demixing crossover between specialized and mixed transcription factories.

Discussion and conclusions

In summary, in this paper we have used coarse-grained simulations to study the 3D structure of human chromatin, its transcriptional dynamics, and their mutual relationship. Unlike previous works [11, 27], here we adopt a multicolour model, viz a polymer model in which chromatin interacts with different types (colours) of complexes between polymerases and chromatin-binding transcription factors (TF:pol). This accounts for the important biological fact that most eukaryotic cells show different kinds of RNA polymerases and a variety of chromatin-binding proteins, with different biological scopes. Our model yields a number of experimentally relevant results.

First, we characterise the morphology of transcription factories (or clusters), arising in our model through the bridging-induced attraction [26]. When these clusters are small, they typically contain TFs of just one colour; these are reminiscent of the specialized transcription factories found in the nucleoplasm that contain active forms of just pol II or pol III – but not both [67]. Instead, when factories are large, they are typically mixed (Fig. 7C); this provides a mechanistic basis for the formation of HOTs, where many different TFs bind promiscuously and weakly to segments of open chromatin that are often devoid of high-affinity specific binding sites [35–37]. The existence of a transition (more precisely, a crossover) between demixed and mixed clusters dependent on cluster size is robust to changes in TF:pol size (Fig. S9), chromatin density (Fig. S10) and the inclusion of active process such as loop extrusion, incorporated through the HiP-HoP model (Fig. 8). The latter simulations also show that the existence of a demixing transition (and the critical size threshold) is not affected by other structurally important ingredients in the model such as the presence of silence or inactive chromatin, and chromatin heteromorphism [41]. This confirms that the mechanism behind the transition is the shift in balance between non-specific and specific chromatin-protein interactions: the former becomes more dominant as cluster size increases. Interestingly, the mechanisms that determine whether a gene belongs to a specialised or mixed factory remain unclear [68]. However, our results suggest that the TF cluster size, along with the 1D TU patterning along the chromatin filament, plays a crucial role, as it links 3D chromatin structure, transcription factory morphology, and gene expression. Specialised and mixed factories thus emerge naturally from TUs arrangement, without the need for additional ingredients, such as posttranscriptional biochemical regulation.

We propose that the predicted demixing–mixing crossover may be experimentally testable in the future through techniques capable of detecting multi-way chromatin interactions, such as SPRITE and GAM [2]. At present, however, these methods have primarily enabled the detection of three-way chromatin contacts [69–72], and statistical data on higher-order interactions remain limited or unavailable [73]. Nevertheless, it is reasonable to expect that ongoing advancements in these technologies will yield increasingly comprehensive datasets, potentially allowing for direct experimental validation of our predictions.

Second, we see remarkable positive and negative correlations in the transcriptional activities of different TUs. For example, activities of same-colour and nearby TUs tend to be strongly positively correlated, as such TUs tend to co-cluster (Fig. 5). Conversely, activities of similar TUs lying far from each other on the genetic map are often weakly negatively correlated, as the formation of one cluster sequesters some TFs to reduce the number available to bind elsewhere (Figure S12).

Taken together, these results provide simple explanations of why adjacent TUs throughout large domains tend to be co-transcribed so frequently [63], as they are likely to gather together in the same cluster. Results also show how one eQTL might up-regulate some TUs and down-regulate

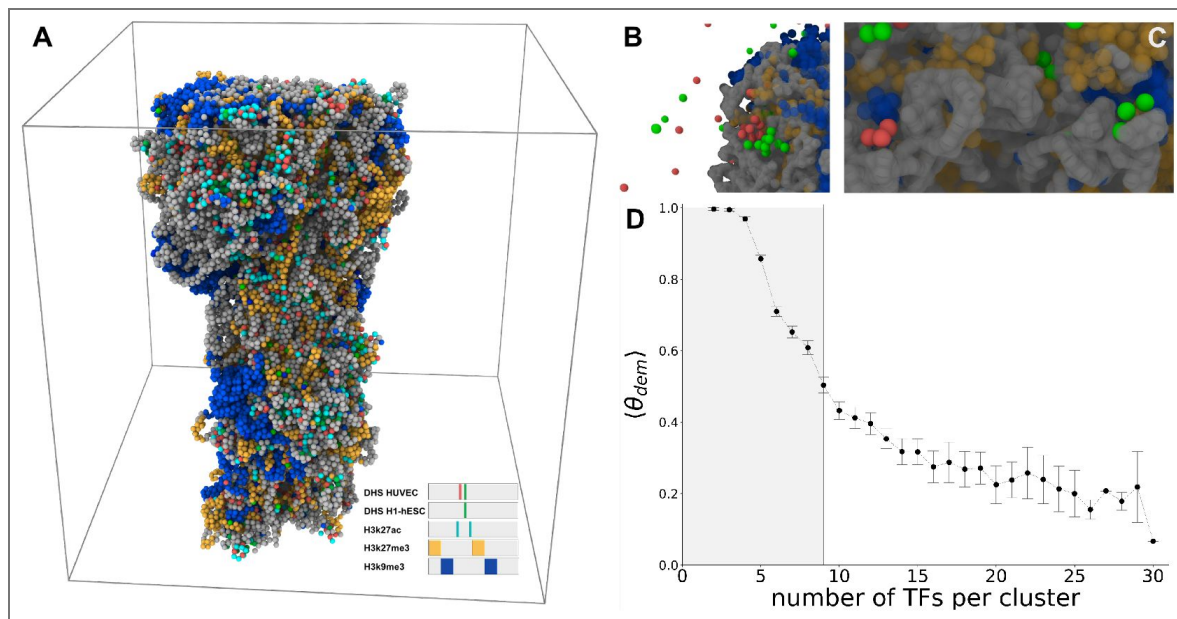


Figure 8. HiP-HoP model simulations: small clusters tend to be unmixed, large ones mixed.

(A) Snapshot of a configuration adopted by HSA14 in HUVECs, within the HiP-HoP model. Grey regions represent less accessible chromatin regions poor in H3K27ac, while cyan regions represent those enriched in H3K27ac. In addition, H3K27me3 and H3K9me3 peaks determine the chromatin binding sites for polycomb-like and heterochromatin proteins, and are represented in yellow and blue respectively. As in the previous DHS multicolour model, TUs only present in HUVEC are represented in red, while the house-keeping ones in green. (B-C) Example of clusters of proteins: large mixed cluster (B) and a small demixed one (C). (D) Average of the demixing coefficient θ_{dem} (error bars: SD). Values of 1 and 0 correspond to completely demixed and completely mixed clusters respectively. Grey area: demixed regime where θ_{dem} is > 0.5 .

others, that can lead to genetic effects like “transgressive segregation” [59].

Third, we can predict effects of local mutations and genome edits that often induce distant omnigenic effects uncovered by genome-wide association studies [11, 58]. For example, mutations that switch a binding site of one TF to another can convert a cluster of one colour into another (Fig. 2 [↗](#)). Similarly, global effects of knocking down TF levels are easily assessed (Fig. 3 [↗](#)).

Fourth, we also predict transcriptional activities of all TUs (both genic and non-genic) on whole human chromosomes by including cell-type-invariant and cell-type-specific TFs (Fig. 6 [↗](#)). We find this yields a better correlation with GRO-seq experimental data than a singlecolour model (where just one TF binds to all TUs similarly). This result underscores the importance of including different TFs in polymer models.

Finally, all our results point to the importance of the 1D pattern of TUs and TF-binding sites on chromosomes in determining activity. In other words, 1D location is a key feature determining transcriptional patterns, and so cell identity. We speculate this is why relative locations of active regulatory elements are so highly conserved. For instance, despite human enhancers evolving much more rapidly than their target protein-coding genes, the synteny between the two (over distances up to 2 Mbp) is highly conserved [74, 75].

In the future, it would be interesting to incorporate the effects of transcription elongation into our model. Although the current implementations of both the simplified toy model and the more detailed HiP-HoP model show good agreement with GRO-seq data, further investigation is needed to assess how molecular motors—such as RNA polymerases—and interactions between RNA molecules and nuclear proteins, including SAF-A [76, 77], contribute to the three-dimensional organization of chromatin. Additionally, it would be valuable to consider the effects of hydrodynamic interactions [78–80]. In our model, as in most chromatin polymer models in the literature [66], hydrodynamic interactions and the resulting spatiotemporal correlations are neglected. Whilst this choice provides significant computational advantages, it also represents a limitation. Although passive hydrodynamic flow associated with polymer motion is likely screened inside the nucleus, the dipolar forces exerted by molecular motors may be strong enough to induce ordering in the intranuclear polymer melt [81]. In fact, recent experiments using Displacement Correlation Spectroscopy, used to map chromatin movements throughout the nucleus in live cells, revealed that chromatin exhibits rapid, uncorrelated motions at short timescales and slower, correlated motions over $\sim\mu\text{m}$ domains at longer timescales [81]. While the typical sizes of the emerging clusters we observe is about an order of magnitude smaller than that of these domains, incorporating hydrodynamics would help elucidate the effect of coherent chromatin dynamics on cluster formation and coarsening.

In addition, it would be of interest to extend the results presented here to incorporate many more types of TFs and TUs within the framework of HiP-HoP [41], and to include dynamic epigenetic modifications [24, 82, 83]. From a theoretical point of view, we hope our results will stimulate the development of theories to understand the mixing-demixing crossover more fundamentally from a polymer physics view-point, as well as more work on the interrelations between 3D structure and function in chromosomes.

Methods

Chromatin fibres are modelled as bead-and-spring polymers [11, 13, 21, 23, 26, 27, 40–47], where each monomer (diameter σ) represents 3 kbp packed into a 30 nm sphere [11, 26, 45]. Different TFs (or TF:pol complexes) are modelled as differently-coloured spheres (also with diameter σ) able to bind (when in an “on” state) to cognate sites of the same colour that are scattered along the polymer. TF:pol complexes and TUs have the same size, since the former represents both transcription factors and polymerases, which in human cells are about 5 nm and 25 nm respectively. We also simulated the case in which TF:pol size is smaller (0.5σ and 0.16σ , Fig. S9) to explore the potential effect of protein size.

Each TF and TF:pol switches between “off” (nonbinding) and “on” (binding) states to reflect the posttranslational modifications that occur in many TFs. Polymer beads are either non-binding (“heterochromatic”), weakly-binding (“euchromatic”), or stronglybinding (containing cognate sites). TFs bind nonspecifically to all weakly-binding beads, and strongly only to TUs of the same colour. TUs in our model represents regulatory elements such as promoters and enhancers, and as discussed below in practice can be identified with DNase hypersensitive regions, which are very sticky for a wide range of TFs, or active protein complexes [84].

The system evolves in a cubic simulation domain with periodic boundary conditions through constanttemperature Langevin dynamics that are integrated numerically by the LAMMPS simulation package [85].

In our model, as in most chromatin polymer models in the literature [1], hydrodynamic interactions are neglected. While this choice offers significant computational advantages, it also presents a limitation. Although passive hydrodynamic flow associated with polymer motion is likely screened inside the nucleus, dipolar forces exerted by molecular motors may still be strong enough to induce ordering in the intranuclear polymer melt, as discussed in [81](see also the Discussion section for additional comments on the possible effects of hydrodynamics). Averages are evaluated over 100 independent runs for each case. The TF volume fraction of each colour is set to $\sim 3 \cdot 10^{-5}$, and the polymer volume fraction to $\sim 2 \cdot 10^{-3}$. We note though that the key control parameter is the ratio between the number of TFs and that of TUs, for each colour. More information about the model can be found in the Supplemental Information (SI). Several quantities are monitored to describe the system’s behavior. Mean transcriptional activity is measured as the fraction of time that a TU is “transcriptionally active”, i.e. within a fixed distance of a TF, in 100 simulations, and so represents a population average (each simulation run may be thought of as a different cell). In order to be quite conservative about the evaluation of the transcriptional activity for TUs in clusters, we adopt 2.25σ as transcriptional threshold. Our point of view is in fact to consider transcriptionally active all TUs in clusters, even when TFs are a bit further. However, we also checked that lower distance thresholds do not affect evaluation of transcriptional profiles in any significant way (see Fig.S11 in SI). Transcriptional activity is then compared with experimental data on transcriptional activity, obtained via GRO-seq – a method providing a genome-wide average readout of ongoing transcription of both genic and non-genic TUs in cell populations [86, 87]. The mean transcriptional Pearson correlation between all pairs of TUs is also evaluated, and a graphical overview of this feature is provided via the Pearson correlation matrix. We also analyse clusters/factories of bound and spatially-proximate TFs, count the number of TFs of similar colour in each cluster, and introduce a demixing coefficient

$$\theta_{\text{dem}} = \frac{nx_{i,\text{max}} - 1}{n - 1}, \quad (1)$$

where n is the number of colors, i is an index denoting each of the colors present in the model and $x_{i,\text{max}}$ the largest fraction of TFs of the same i -th color in a single TF cluster. If $\theta_{\text{dem}} = 1$, this means that a cluster contains only TFs of one colour and so is fully demixed; if $\theta_{\text{dem}} = 0$, the cluster contains a mixture of TFs of all colors in equal number, and so is maximally demixed. More details can be found in the SI.

We consider two different types of string, one with $M = 3000$ beads (or 9 Mbp) which is referred to as a “toy” string, and a second representing a whole human chromosome. Chromosomes are initialised in both cases as random walks. An alternative possibility would be to start from mitotic configuration as in [88], which would remove entanglement in the initial condition. Experience with similar models (e.g., see [50]) suggests that a different initial condition will be important for the very large-scale structure but not for the scale at which transcriptional clusters form, which is the one we are most interested in here.

Toy model

The toy model is built by placing one yellow, red, or green TU every 30 weakly-binding beads, giving a total of 100 TUs of all types in a string of 3000 beads [26]. Various different sequences of TU colour down the string are considered. In one – the “random” string – TU colours are chosen randomly (see Fig. 1a [26] and SI for the specific sequence generated). In a second and third – the “1-pattern” and “6-pattern” strings – TU colors follow a repeating pattern (red, then yellow, then green) 1 or 6 times (see Fig. 4 [26]). We made these choices for the sequences of TUs as they are useful to show how 1D patterns affect resulting cluster morphology. In this respect, these patterns in the toy model are only representative. At the same time, the ratio between TFs and TUs are close to those used below for human chromosome simulations.

For the random string, we monitor how the system responds to different perturbations. Local “mutations” are inspired by editing experiments performed using CRISPR/Cas9 [60]. One to four mutations are mimicked by switching selected yellow beads inside a cluster of consecutive yellow TUs (between TUs 1920 to 2070) to red ones (Fig. 2 [26]). Thus, conversion of TU bead 1980 gives a string with 1 mutation, of 1950 and 1980 gives 2 mutations, of 1950 to 2010 gives 3 mutations, and 1950 to 2040 gives 4 mutations. Global perturbations are inspired by experiments reducing global levels of TFs using auxin-induced degrons [61]. Here, we study the effects of reducing the concentration of yellow TFs by 30%.

Human chromosomes

Our reference case for whole human chromosome simulations in the main text is the mid-sized human chromosome HSA 14 (107 Mbp), coarse-grained into $M = 35784$ beads. For Fig. 6 [26], weakly- and strongly-binding beads are identified (using ENCODE data [84] for human umbilical vein endothelial cells, HUVECs) by the presence of H3K27ac modifications and DNase-hypersensitivity sites (DHSs) in the 3 kbp region corresponding to that bead – as these are good markers of open chromatin and active TUs (both genic and non-genic), respectively. For Fig. 6 [26], TUs are split into ones only active in HUVECs and others (“house-keeping” ones) that are also active in H1-hESC cells (again using DHS sites and ENCODE data). Then, if a TU appears in both HUVECs and H1-hESCs, it is marked as housekeeping and coloured red; if it appears only in HUVECs it is marked as HUVEC-specific and coloured green. This allows an intuitive and simple multicolour model of HUVECs to be constructed. All remaining beads (which are not either weakly-binding or TUs) are non-binding. This approach represents a generalisation of the DHS model described in [11], so we call it the multicolour DHS model. For the simulations shown in the main text TF:pol complexes and TU size is the same (σ , corresponding to 30 nm at our resolution). This is justified by the fact that our TF:pol represents both transcription factors and polymerases. A polymerase is about 25 nm in human cells [30], while transcription factors are typically at least 5 nm in size. We also considered the case in which TF:pol size is smaller (0.5σ and 0.16σ , Fig. S9) to explore the potential effect of protein size: as we shall see, this does not qualitatively affect our conclusions and results.

We also consider HSA 18 (80 Mbp, 26026 beads) and 19 (58 Mbp, 19710 beads) in HUVECs, chosen as they represent gene-poor and gene-rich chromosomes, respectively. Additionally, we consider HSA 14 in the B-lymphocyte line GM12878 (again, colours are chosen by combining DHS data for GM12878 and H1-hESCs). H3K27ac and DHS data is again from ENCODE.

The multicolor DHS model was also applied within a more realistic chromatin framework, the “highly predictive heteromorphic polymer model”, or HiP-HoP model [17]. This is a much more sophisticated model which takes into account: (i) loop extrusion; (ii) inactive (as well as active) chromatin folding; (iii) chromatin heteromorphism (different local compaction of chromatin according to acetylation). More details on the HiP-HoP model are given in the SI.

For human chromosomes, transcriptional-activity data obtained from simulations and GRO-seq are compared in two ways [11]. First, we rank activities of each TU, and build a two-dimensional histogram providing an overview of the agreement between the two sets of ranks. Second, we quantify Spearman’s rank correlation coefficient between numerical and experimental data (SI for more details).

Data availability

All experimental data used in this paper are available in the ENCODE database [84]. All custom scripts used for the simulations presented here are available in the Zenodo database 10.5281/zenodo.17408464.

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Additional files

[Supplemental Material](#) 

Additional information

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Peer reviews

Reviewer #1 (Public review):

This manuscript discusses from a theory point of view the mechanisms underlying the formation of specialized or mixed factories. To investigate this, a chromatin polymer model was developed to mimic the chromatin binding-unbinding dynamics of various complexes of transcription factors (TFs).

The model revealed that both specialized (i.e., demixed) and mixed clusters can emerge spontaneously, with the type of cluster formed primarily determined by cluster size. Non-specific interactions between chromatin and proteins were identified as the main factor promoting mixing, with these interactions becoming increasingly significant as clusters grow larger.

These findings, observed in both simple polymer models and more realistic representations of human chromosomes, reconcile previously conflicting experimental results. Additionally, the introduction of different types of TFs was shown to strongly influence the emergence of transcriptional networks, offering a framework to study transcriptional changes resulting from gene editing or naturally occurring mutations.

Overall I think this is an interesting paper discussing a valuable model of how chromosome 3D organisation is linked to transcription.

Comments on revisions: It's a good paper.

<https://doi.org/10.7554/eLife.103955.2.sa3>

Reviewer #2 (Public review):

Summary:

With this report, I suggest what are in my opinion crucial additions to the otherwise very interesting and credible research manuscript "Cluster size determines morphology of transcription factories in human cells".

Strengths:

The manuscript in itself is technically sound, the chosen simulation methods are completely appropriate the figures are well-prepared, the text is mostly well-written spare a few typos. The conclusions are valid and would represent a valuable conceptual contribution to the field of clustering, 3D genome organization and gene regulation related to transcription factories, which continues to be an area of most active investigation.

Weaknesses:

However, I find that the connection to concrete biological data is weak. This holds especially given that the data that are needed to critically assess the applicability of the derived cross-over with factory size is, in fact, available for analysis, and the suggested experiments in the Discussion section are actually done and their results can be exploited. In my judgement, unless these additional analysis are added to a level that crucial predictions on TF demixing and transcriptional bursting upon TU clustering can be tested, the paper is more fitted for a theoretical biophysics venue than for a biology journal such as eLife.

Comments on revisions:

The authors have addressed my comments with exemplary diligence, which has clarified all my major concerns. In all cases, either the relevant work was added, or it was explained in the form of a convincing argument why the suggested modifications were not implemented or not possible to implement.

As a discretionary suggestion, the authors might consider using a title that even more directly highlights the, in my opinion, main take-away of this work. This is not because anything is incorrect about the current title, simply an even more to-the-point title might attract more readers. I would suggest something along the lines of

"Cluster size-dependent demixing drives specialization of transcription factories"

Overall, I congratulate the authors on their excellent work and appreciate the opportunity to engage with this manuscript during a very insightful review process.

<https://doi.org/10.7554/eLife.103955.2.sa2>

Reviewer #3 (Public review):

Summary:

In this work, the authors present a chromatin polymer model with some specific pattern of transcription units (TUs) and diffusing TFs; they simulate the model and study TFclustering, mixing, gene expression activity, and their correlations. First, the authors designed a toy polymer with colored beads of a random type, placed periodically (every 30 beads, or 90kb). These colored beads are considered a transcription unit (TU). Same-colored TUs attract with each other mediated by similarly colored diffusing beads considered as TFs. This led to clustering (condensation of beads) and correlated (or anti-correlation) "gene expression" patterns. Beyond the toy model, when authors introduce TUs in a specific pattern, it leads to emergence of specialized and mixed cluster of different TFs. Human chromatin models with realistic distribution of TUs also lead to the mixing of TFs when cluster size is large.

Strengths:

This is a valuable polymer model for chromatin with a specific pattern of TUs and diffusing TF-like beads. Simulation of the model tests many interesting ideas. The simulation study is

convincing and the results provide solid evidence showing the emergence of mixed and demixed TF clusters within the assumptions of the model.

<https://doi.org/10.7554/eLife.103955.2.sa1>

Author response:

The following is the authors' response to the original reviews

eLife Assessment

This is a valuable polymer model that provides insight into the origin of macromolecular mixed and demixed states within transcription clusters. The well-performed and clearly presented simulations will be of interest to those studying gene expression in the context of chromatin. While the study is generally solid, it could benefit from a more direct comparison with existing experimental data sets as well as further discussion of the limits of the underlying model assumptions.

We thank the editors for their overall positive assessment. In response to the Referees' comments, we have addressed all technical points, including a more detailed explanation of the methodology used to extract gene transcription from our simulations and its analogy with real gene transcription. Regarding the potential comparison with experimental data and our mixing–demixing transition, we have added new sections discussing the current state of the art in relevant experiments. We also clarify the present limitations that prevent direct comparisons, which we hope can be overcome with future experiments using the emerging techniques.

Reviewer #1 (Public Review):

This manuscript discusses from a theory point of view the mechanisms underlying the formation of specialized or mixed factories. To investigate this, a chromatin polymer model was developed to mimic the chromatin binding-unbinding dynamics of various complexes of transcription factors (TFs).

The model revealed that both specialized (i.e., demixed) and mixed clusters can emerge spontaneously, with the type of cluster formed primarily determined by cluster size. Non-specific interactions between chromatin and proteins were identified as the main factor promoting mixing, with these interactions becoming increasingly significant as clusters grow larger.

These findings, observed in both simple polymer models and more realistic representations of human chromosomes, reconcile previously conflicting experimental results. Additionally, the introduction of different types of TFs was shown to strongly influence the emergence of transcriptional networks, offering a framework to study transcriptional changes resulting from gene editing or naturally occurring mutations.

Overall I think this is an interesting paper discussing a valuable model of how chromosome 3D organisation is linked to transcription. I would only advise the authors to polish and shorten their text to better highlight their key findings and make it more accessible to the reader.

We thank the Referee for carefully reading our manuscript and recognizing its scientific value. As suggested, we tried to better highlight our key findings and make the text more accessible while addressing also the comments from the other Referees.

Reviewer #2 (Public Review):

Summary:

With this report, I suggest what are in my opinion crucial additions to the otherwise very interesting and credible research manuscript "Cluster size determines morphology of transcription factories in human cells".

Strengths:

The manuscript in itself is technically sound, the chosen simulation methods are completely appropriate the figures are well-prepared, the text is mostly well-written spare a few typos. The conclusions are valid and would represent a valuable conceptual contribution to the field of clustering, 3D genome organization and gene regulation related to transcription factories, which continues to be an area of most active investigation.

Weaknesses:

However, I find that the connection to concrete biological data is weak. This holds especially given that the data that are needed to critically assess the applicability of the derived cross-over with factory size is, in fact, available for analysis, and the suggested experiments in the Discussion section are actually done and their results can be exploited. In my judgement, unless these additional analysis are added to a level that crucial predictions on TF demixing and transcriptional bursting upon TU clustering can be tested, the paper is more fitted for a theoretical biophysics venue than for a biology journal such as eLife.

We thank the Reviewer for their positive assessment of the soundness of our work and its contribution to the field. We have added a paragraph to the Conclusions highlighting the current state of experimental techniques and outlining near-term experiments that could be extended to test our predictions. We also emphasise that our analysis builds on state-of-the-art polymer models of chromatin and on quantitative experimental datasets, which we used both to build the model construction and to validate its outcomes (gene activity). We hope this strengthened link to experiment will catalyse further studies in the field.

Major points:

(1) My first point concerns terminology. The Merriam-Webster dictionary describes morphology as the study of structure and form. In my understanding, none of the analyses carried out in this study actually address the form or spatial structuring of transcription factories. I see no aspects of shape, only size. Unless the authors want to assess actual shapes of clusters, I would recommend to instead talk about only their size/extent. The title is, by the same argument, in my opinion misleading as to the content of this study.

We agree with the Referee that the title could be misleading. In our study we characterized clusters size, that is a morphological descriptor, and cluster composition that isn't morphology per se but used in the community in a broader sense. Nevertheless to strengthen the message we have changed the title in: "Cluster size determines internal structure of transcription factories in human cells"

(2) Another major conceptual point is the choice of how a single TF:pol particle in the model relates to actual macromolecules that undergo clustering in the cell. What about the fact that even single TF factories still contain numerous canonical transcription factors, many of which are also known to undergo phase separation? Mediator, CDK9, Pol II just to name a few. This alone already represents phase separation under the involvement of different species, which must undergo mixing. This is conceptually blurred with the concept of gene-specific transcription factors that are recruited into clusters/condensates due to sequencespecific or chromatin-epigenetic-specific affinities.

Also, the fact that even in a canonical gene with a “small” transcription factory there are numerous clustering factors takes even the smallest factories into a regime of several tens of clustering macromolecules. It is unclear to me how this reality of clustering and factory formation in the biological cell relates to the cross-over that occurs at approximately $n=10$ particles in the simulations presented in this paper.

This is a good point. However in our case we can either look at clustering transcription factors or transcription units. In an experimental situation, transcription units could be “coloured”, or assigned different types, by looking at different cell types, so that they can be classified as housekeeping, or cell-type independent, or cell-type specific. This is similar to how DHS can be clustered. In this way the mixing or demixing state can be identified by looking at the type of transcription unit, removing any ambiguity due to the fact that the same protein may participate in different TF complexes..

(3) The paper falls critically short in referencing and exploiting for analysis existing literature and published data both on 3D genome organization as well as the process of cluster formation in relation to genomic elements. In terms of relevant literature, most of the relevant body of work from the following areas has not been included:

(i) mechanisms of how the clustering of Pol II, canonical TFs, and specific TFs is aided by sequence elements and specific chromatin states

(ii) mechanisms of TF selectivity for specific condensates and target genomic elements

(iii) most crucially, existing highly relevant datasets that connect 3D multi-point contacts with transcription factor identity and transcriptional activity, which would allow the authors to directly test their hypotheses by analysis of existing data

Here, especially the data under point (iii) are essential. The SPRITE method (cited but not further exploited by the authors), even in its initial form of publication, would have offered a data set to critically test the mixing vs. demixing hypothesis put forward by the authors. Specifically, the SPRITE method offers ordered data on k -mers of associated genomic elements. These can be mapped against the main TFs that associate with these genomic elements, thereby giving an account of the mixed / demixed state of these k -mer associations. Even a simple analysis sorting these associations by the number of associated genomic elements might reveal a demixing transition with increasing association size k . However, a newer version of the SPRITE method already exists, which combines the k -mer association of genomic elements with the whole transcriptome assessment of RNAs associated with a particular DNA k -mer association. This can even directly test the hypotheses the authors put forward regarding cluster size, transcriptional activation, correlation between different transcription units' activation etc.

To continue, the Genome Architecture Mapping (GAM) method from Ana Pombo's group has also yielded data sets that connect the long-range contacts between gene-regulatory elements to the TF motifs involved in these motifs, and even provides ready-made analyses that assess how mixed or demixed the TF composition at different interaction hubs is. I do not see why this work and data set is not even acknowledged? I also strongly suggest to analyze, or if they are already sufficiently analyzed, discuss these data in the light of 3D interaction hub size (number of interacting elements) and TF motif composition of the involved genomic elements.

Further, a preprint from the Alistair Boettiger and Kevin Wang labs from May 2024 also provides direct, single-cell imaging data of all super-enhancers, combined with transcription detection, assessing even directly the role of number of super-enhancers in spatial proximity as a determinant of transcriptional state. This data set and findings

should be discussed, not in vague terms but in detailed terms of what parts of the authors' predictions match or do not match these data.

For these data sets, an analysis in terms of the authors' key predictions must be carried out (unless the underlying papers already provide such final analysis results). In answering this comment, what matters to me is not that the authors follow my suggestions to the letter. Rather, I would want to see that the wealth of available biological data and knowledge that connects to their predictions is used to their full potential in terms of rejecting, confirming, refining, or putting into real biological context the model predictions made in this study.

References for point (iii):

- RNA promotes the formation of spatial compartments in the nucleus
[https://www.cell.com/cell/fulltext/S0092-8674\(21\)01230-7?dgcid=raven_jbs_etoc_email](https://www.cell.com/cell/fulltext/S0092-8674(21)01230-7?dgcid=raven_jbs_etoc_email)
- Complex multi-enhancer contacts captured by genome architecture mapping
<https://www.nature.com/articles/nature21411>
- Cell-type specialization is encoded by specific chromatin topologies
<https://www.nature.com/articles/s41586-021-04081-2>
- Super-enhancer interactomes from single cells link clustering and transcription
<https://www.biorxiv.org/content/10.1101/2024.05.08.593251v1.full>

For point (i) and point (ii), the authors should go through the relevant literature on Pol II and TF clustering, how this connects to genomic features that support the cluster formation, and also the recent literature on TF specificity. On the last point, TF specificity, especially the groups of Ben Sabari and Mustafa Mirx have presented astonishing results, that seem highly relevant to the Discussion of this manuscript.

We appreciate the Reviewer's insightful suggestion that a comparison between our simulation results and experimental data would strengthen the robustness of our model. In response, we have thoroughly revised the literature on multi-way chromatin contacts, with particular attention to SPRITE and GAM techniques. However, we found that the currently available experimental datasets lack sufficient statistical power to provide a definitive test of our simulation predictions, as detailed below.

As noted by the Reviewer, SPRITE experiments offer valuable information on the composition of highorder chromatin clusters (k-mers) that involve multiple genomic loci. A closer examination of the SPRITE data (e.g., Supplementary Material from Ref. [1]) reveals that the majority of reported statistics correspond to 3-mers (three-way contacts), while data on larger clusters (e.g., 8-mers, 9-mers, or greater) are sparse. This limitation hinders our ability to test the demixing-mixing transition predicted in our simulations, which occurs for cluster sizes exceeding 10.

Moreover, the composition of the k-mers identified by SPRITE predominantly involves genomic regions encoding functional RNAs—such as ITS1 and ITS2 (involved in rRNA synthesis) and U3 (encoding small nucleolar RNA)—which largely correspond to housekeeping genes. Conversely, there is little to no data available for protein-coding genes. This restricts direct comparison to our simulations, where the demixing-mixing transition depends critically on the interplay between housekeeping and tissue-specific genes.

Similarly, while GAM experiments are capable of detecting multi-way chromatin contacts, the currently available datasets primarily report three-way interactions [2,3].

In summary, due to the limited statistical data on higher-order chromatin clusters [4], a quantitative comparison between our simulation results and experimental observations is

not currently feasible. Nevertheless, we have now briefly discussed the experimental techniques for detecting multi-way interactions in the revised manuscript to reflect the current state of the field, mentioning most of the references that the Reviewer suggested.

(4) Another conceptual point that is a critical omission is the clarification that there are, in fact, known large vs. small transcription factories, or transcriptional clusters, which are specific to stem cells and “stressed cells”. This distinction was initially established by Ibrahim Cisse’s lab (Science 2018) in mouse Embryonic Stem Cells, and also is seen in two other cases in differentiated cells in response to serum stimulus and in early embryonic development:

- Mediator and RNA polymerase II clusters associate in transcription-dependent condensates <https://www.science.org/doi/10.1126/science.aar4199>

- Nuclear actin regulates inducible transcription by enhancing RNA polymerase II clustering <https://www.science.org/doi/10.1126/sciadv.aay6515>

- RNA polymerase II clusters form in line with surface condensation on regulatory chromatin <https://www.embopress.org/doi/full/10.15252/msb.202110272>

- If “morphology” should indeed be discussed, the last paper is a good starting point, especially in combination with this additional paper: Chromatin expansion microscopy reveals nanoscale organization of transcription and chromatin <https://www.science.org/doi/10.1126/science.ade5308>

We thank the Reviewer for pointing out the discussion about small and large clusters observed in stressed cells. Our study aims to provide a broader mechanistic explanation on the formation of TF mixed and demixed clusters depending on their size. However, to avoid to generate confusion between our terminology and the classification that is already used for transcription factories in stem and stressed cells, we have now added some comments and references in the revised text.

(5) The statement scripts are available upon request is insufficient by current FAIR standards and seems to be non-compliant with eLife requirements. At a minimum, all, and I mean all, scripts that are needed to produce the simulation outcomes and figures in the paper, must be deposited as a publicly accessible Supplement with the article. Better would be if they would be structured and sufficiently documented and then deposited in external repositories that are appropriate for the sharing of such program code and models.

We fully agree with the Reviewer. We have now included in the main text a link to an external repository containing all the codes required to reproduce and analyze the simulations.

Recommendations for the authors:

Minor and technical points

(6) Red, green, and yellow (mix of green and red) is a particularly bad choice of color code, seeing that red-green blindness is the most common color blindness. I recommend to change the color code.

We appreciate the Reviewer’s thoughtful comment regarding color accessibility. We fully agree that red–green combinations can pose challenges for color-blind readers. In our figures, however, we chose the red–green–yellow color scheme deliberately because it provides strong contrast and intuitive representation for different TF/TU types. To ensure accessibility, we optimized brightness and saturation within red–green schemes and we carefully verified that the chosen hues are distinguishable under the most common forms of

color vision deficiency, i.e. trichromatic color blindness, using color-blindness simulation tools (e.g., Coblis).

How is the dispersing effect of transcriptional activation and ongoing transcription accounted for or expected to affect the model outcome? This affects both transcriptional clusters (they tend to disintegrate upon transcriptional activation) as well as the large scale organization, where dispersal by transcription is also known.

We thank the Reviewer for this very insightful question. The current versions of both our toy model and the more complex HiP-HoP model do not incorporate the effects of RNA Polymerase elongation. Our primary goal was to develop a minimalistic framework that focuses on investigating TF clusters formation and their composition. Nevertheless, we find that this straightforward approach provides a good agreement between simulations and Hi-C and GRO-seq experiments, lending confidence to the reliability of our results concerning TF cluster composition.

We fully agree, however, that the effects of transcription elongation are an interesting topic for further exploration. For example, modeling RNA Polymerases as active motors that continually drive the system out of equilibrium could influence the chromatin polymer conformation and the structure of TF clusters. Additionally, investigating how interactions between RNA molecules and nuclear proteins, such as SAF-A, might lead to significant changes in 3D chromatin organization and, consequently, transcription [5], is also an intriguing prospect. Although we do not believe that the main findings of our study, particularly regarding cluster composition and mixed-demixed transition, would be impacted by transcription elongation effects, we recognize the importance of this aspect. As such, we have now included some comments in the Conclusions section of the revised manuscript.

“and make the reasonable assumption that a TU bead is transcribed if it lies within 2.25 diameters (2.25σ) of a complex of the same colour; then, the transcriptional activity of each TU is given by the fraction of time that the TU and a TF:pol lie close together.” How is that justified? I do not see how this is reasonable or not, if you make that statement you must back it up.

As pointed out by the Referee, we consider a TU to be active if at least one TF is within a distance 2.25σ from that TU. This threshold is a slightly larger than the TU-TF interaction cutoff distance, $r_c = 1.8\sigma$ between TFs and TUs. The rationale for this choice is to ensure that, in the presence of a TU cluster surrounded by TFs, TUs that are not directly in contact with a TF are still considered active. Nonetheless, we find that using slightly different thresholds, such as 1.8σ or 1.1σ , leads to comparable results, as shown in Fig. S11, demonstrating the robustness of our analysis.

Clearly, close proximity in 1D genomic space favours formation of similarly-coloured clusters. This is not surprising, it is what you built the model to do. Should not be presented as a new insight, but rather as a check that the model does what is expected.

We believed that this sentence already conveyed that the formation of single-color clusters driven by 1D genomic proximity is not a surprising outcome. However, we have now slightly rephrased it to better emphasize that this is not a novel insight.

That said, we would like to highlight that while 1D genomic proximity facilitates the formation of clusters of the same color, the unmixed-to-mixed transition in cluster composition is not easily predictable solely from the TU color pattern. Furthermore, in simulations of real chromosomes, where TU patterns are dictated by epigenetic marks, the complexity of these patterns makes it challenging—if not impossible—to predict cluster composition based solely on the input data of our model.

“...how closely transcriptional activities of different TUs correlate...” Please briefly state over what variable the correlation is carried out, is it cross correlation of transcription activity time courses over time? Would be nice to state here directly in the main text to make it easier for the reader.

We have now included a brief description in the revised manuscript explaining how the transcriptional correlations were evaluated and how the correlation matrix was constructed.

“The second concerns how expression quantitative trait loci (eQTLs) work. Current models see them doing so post-transcriptionally in highly-convoluted ways [11, 55], but we have argued that any TU can act as an eQTL directly at the transcriptional level [11].” This text does not actually explain what eQTLs do. I think it should, in concise words.

We agree with the Referee’s suggestion. We have revised the sentence accordingly and now provide a clear explanation of eQTLs upon their first mention. The revised paragraph now reads as follows:

“The second concerns how expression quantitative trait loci (eQTLs)—genomic regions that are statistically associated with variation in gene expression levels—function. While current models often attribute their effects to post-transcriptional regulation through complex mechanisms [6,7], we have previously argued that any transcriptional unit (TU) can act as an eQTL by directly influencing gene expression at the transcriptional level [7]. Here, we observe individual TUs up-regulating or down-regulating the activity of others TUs – hallmark behaviors of eQTLs that can give rise to genetic effects such as “transgressive segregation” [8]. This phenomenon refers to cases in which alleles exhibit significantly higher or lower expression of a target gene, and can be, for instance, caused by the creation of a non-parental allele with a specific combination of QTLs with opposing effects on the target gene.”

“In the string with 4 mutations, a yellow cluster is never seen; instead, different red clusters appear and disappear (Fig. 2Eii)...” How should it be seen? You mutated away most of the yellow beads. I think the kymograph is more informative about the general model dynamics, not the effects of mutations. Might be more appropriate to place a kymograph in Figure 1.

We agree with the Referee that the kymograph is the most appropriate graphical representation for capturing the effects of mutations. Panel 2E already refers to the standard case shown in Figure 1. We have now clarified this both in the caption and in the main text. In addition, we have rephrased the sentence—which was indeed misleading—as follows:

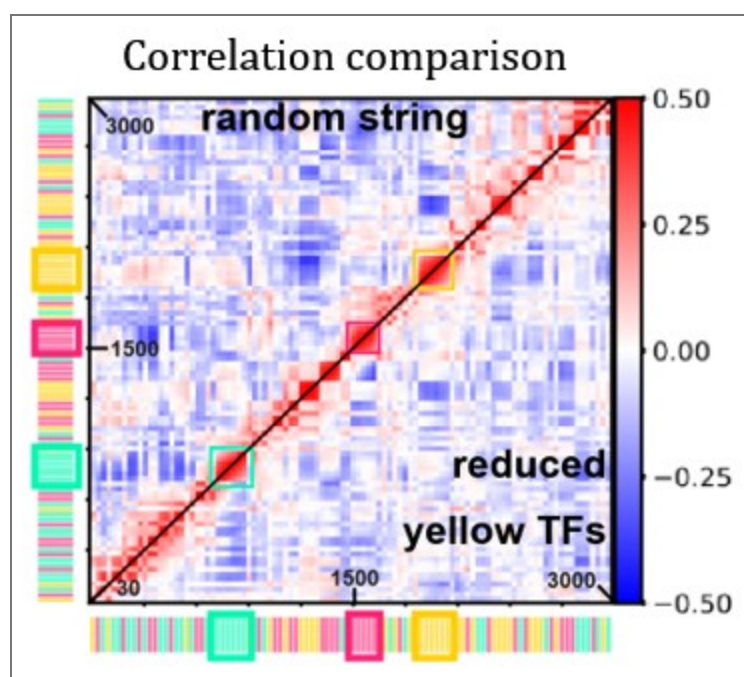
“From the activity profiles in Fig. 2C, we can observe that as the number of mutations increases, the yellow cluster is replaced by a red cluster, with the remaining yellow TUs in the region being expelled (Fig. 2B(ii)). This behavior is reflected in the dynamics, as seen by comparing panels E(i) and E(ii): in the string with four mutations, transcription of the yellow TUs is inhibited in the affected region, while prominent red stripes—corresponding to active, transcribing clusters—emerge (Fig. 2E(ii)).” We hope that the comparison is now immediately clear to the reader.

“...but this block fragments in the string with 4 mutations...” I don’t know or cannot see what is meant by “fragmentation” in the correlation matrix.

With the sentence “this block fragments in the string with 4 mutations” we mean that the majority of the solid red pixels within the black box become light-red or white once the mutations are applied. We have now added a clarification of this point in the revised manuscript.

"Fig. 3D shows the difference in correlation between the case with reduced yellow TFs and the case displayed in Fig. 1E." Can you just place two halves of the different matrices to be compared into the same panel? Similar to Fig. S5. Will be much easier to compare.

We thank the Referee for this suggestion. We tried to implement this modification, and report the modified figure below (Author response image 1). As we can see, in the new figure it is difficult to spot the details we refer to in the main text, therefore we prefer to keep the original version of the figure.



Author response image 1. Heatmap comparing activity correlations of TUs in the random string under normal conditions (top half) and with reduced yellow-TF concentration (bottom half).

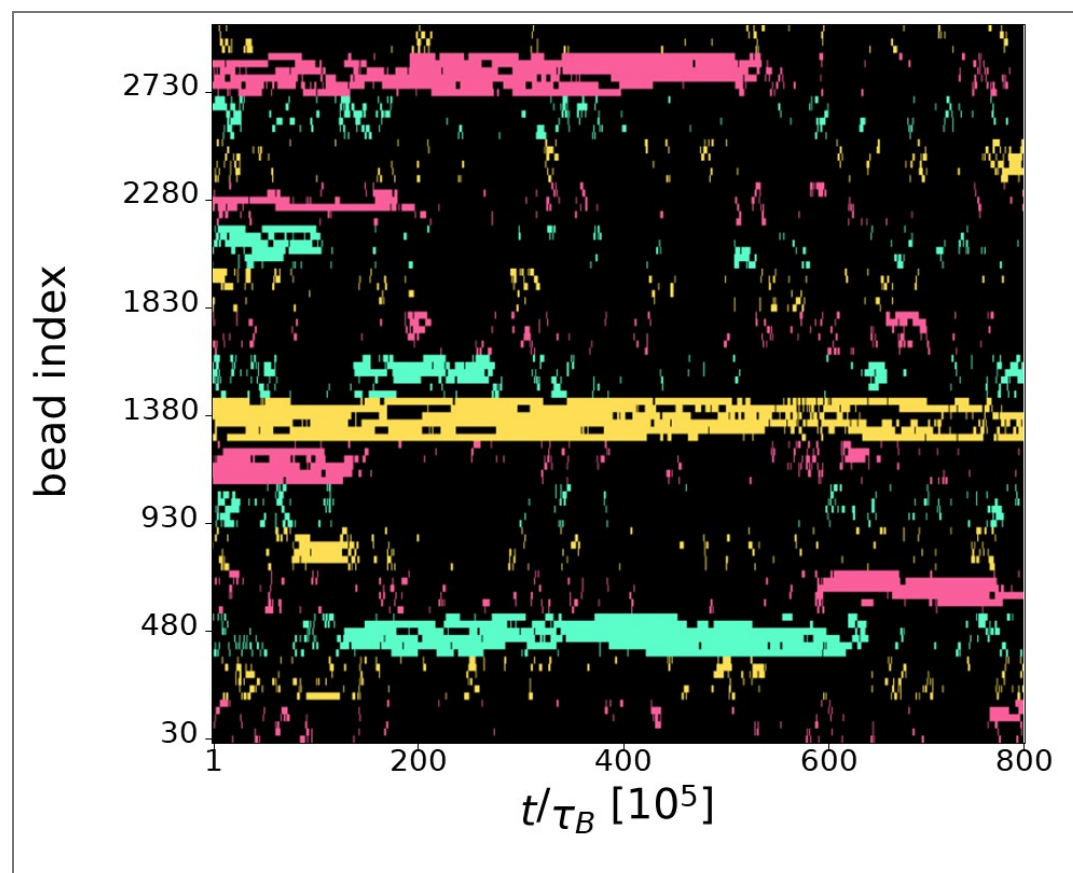
What is the omnigenic model? It is not introduced.

We thank the Reviewer for highlighting this important point. The omnigenic model, first introduced by Boyle et al in Ref. [6], was proposed to explain how complex traits, including disease risk, are influenced by a vast number of genes. Accordingly to this model, the genetic basis of a trait is not limited to a small set of core genes whose expression is directly related to the trait, but also includes peripheral genes. The latter, although not directly involved in controlling the trait, can influence the expression of core genes through gene regulatory networks, thereby contributing to the overall genetic influence on the trait. We have now added a few lines in the revised manuscript to explain this point.

"Additionally, blue off-diagonal blocks indicate repeating negative correlations that reflect the period of the 6-pattern." How does that look in a kymograph? Does this mean the 6 clusters of same color steal the TFs from the other clusters when they form?

The intuition of the Referee is indeed correct. The finite number of TFs leads to competition among TUs of the same colour, resulting in anticorrelation: when a group of six nearby TUs of a given colour is active, other, more distant TUs of the same colour are not transcribing due to the lack of available TFs. As the Referee suggested, this phenomenon is visible in the kymograph showing TU activity. In Author response image 2, it can be observed that typically there is a single TU cluster for each of the three colours (yellow, green, and red). These

clusters can be long-lived (e.g., the yellow cluster at the center of the kymograph) or may destroy during the simulation (e.g., the red cluster at the top of the kymograph, which dissolves at $t \sim 600 \times 10^5 \tau_B$). In the latter case, TFs of the corresponding colour are released into the system and can bind to a different location, forming a new cluster (as seen with the red cluster forming at the bottom of the kymograph for $t > 600 \times 10^5 \tau_B$). This point is further discussed at the point 2.30 of this Reply where additional graphical material is provided.



Author response image 2. Kymograph showing the TU activity during a typical run in the 6-pattern case.

Each row reports the transcriptional state of a TU during one simulation. Black pixels correspond to inactive TUs, red (yellow, green) pixels correspond to active red (yellow, green) TUs.

"Conversely, negative correlations connect distant TUs, as found in the single-color model..." But at the most distal range, the negative correlation is lost again! Why leave this out? Your correlation curves show the same, equilibration towards no correlation at very long ranges.

As highlighted in Figure 5Ai, long-range negative correlations (grey segments) predominantly connect distant TUs of the same colour. This is quantified in Figure 5Bi: restricting to same-colour TUs shows that at large genomic separations the correlation is almost entirely negative, with small fluctuations at distances just below 3000 kbp where sampling is sparse; we therefore avoid further interpretation of this regime.

"These results illustrate how the sequence of TUs on a string can strikingly affect formation of mixed clusters; they also provide an explanation of why activities of human TUs within genomic regions of hundreds of kbp are positively correlated [60]." This is a very nice insight.

We thank the Reviewer for the very supportive comment.

“To quantify the extent to which TFs of different colours share clusters, we introduce a demixing coefficient, θ_{dem} (defined in Fig. 1).” This is not defined in Fig. 1 or anywhere else here in the main text.

We thank the Referee for pointing this out. For a given cluster, the demixing coefficient is defined as

$$\theta_{dem} = \frac{nx_{i,max} - 1}{n - 1}, \quad (1)$$

where n is the number of colors, i indexes each color present in the model, and $x_{i,max}$ the largest fraction of TFs of the same i -th color in a single TF cluster.

The demixing coefficient is defined in the Methods section; therefore, we have replaced defined in Fig. 1 with see Methods for definition.

“Mixing is facilitated by the presence of weakly-binding beads, as replacing them with non-interacting ones increases demixing and reduces long-range negative correlations (Figure S3). Therefore, the sequence of strong and weak binding sites along strings determines the degree of mixing, and the types of small-world network that emerge. If eQTLs also act transcriptionally in the way we suggest [11], we predict that down-regulating eQTLs will lie further away from their targets than up-regulating ones.” Going into these side topics and minke points here is super distracting and waters down the message. Maybe first deal with the main conclusions on mixed vs demixed clusters in dependence on the strong and specific binding site patterns, before dealing with other additional points like the role of weak binding sites.

Thank you for the suggestion. We now changed the paragraph to highlight the main results. The new paragraph is as follows. “These results on activity correlation and TF cluster composition suggest that, if eQTLs act transcriptionally as expected [7], down-regulating eQTLs are likely to be located further from their target genes than up-regulating ones. In addition, it is important to note that mixing is promoted by the presence of weakly binding beads; replacing these with non-interacting ones leads to increased demixing and a reduction in long-range negative correlations (Figure S3). More generally, our findings indicate that the presence of multiple TF colors offers an effective mechanism to enrich and fine-tune transcriptional regulation.”

“...provides a powerful pathway to enrich and modulate transcriptional regulation.” Before going into the possible meaning and implications of the results, please discuss the results themselves first.

See previous point.

Figure 5B. Does activation typically coincide with spatial compaction of the binding sites into a small space or within the confines of a condensate? My guess would be that colocalization of the other color in a small space is what leads to the mixing effect?

As the Reviewer correctly noted, the activity of a given TU is indeed influenced by the presence of nearby TUs of the same color, since their proximity facilitates the recruitment of additional TFs and enhances the overall transcriptional activity. In this context, the mixing effect is certainly affected by the 1D arrangement of TUs along the chromatin fiber. As emphasized in the revised manuscript, when domains of same-color TUs are present (as in the 6-pattern string), the degree of demixing is greater compared to the case where TUs of different colors alternate and large domains are absent (as in the 1-pattern string). This difference in the demixing parameter as a function of the 1D TU arrangement is clearly visible in Fig. S2B.

“...euchromatic regions blue, and heterochromatic ones grey.” Please also explain what these color monomers mean in terms of non specific interactions with the TFs.

Generally, in our simulation approach we assume euchromatin regions to be more open and accessible to transcription factors, whereas heterochromatin corresponds to more compacted chromatin segments [9]. To reflect this, we introduce weak, non-specific interactions between euchromatin and TFs, while heterochromatin interacts with TFs only through steric effects. To clarify this point, we have now slightly revised the caption of Fig.6.

“More quantitatively, Spearman’s rank correlation coefficient is $3.66 \cdot 10^{-1}$, which compares with $3.24 \cdot 10^{-1}$ obtained previously using a single-colour model [11].” This comparison does not tell me whether the improvement in model performance justifies an additional model component. There are other, likelihood based approaches to assess whether a model fits better in a relevant extent by adding a free model parameter. Can these be used for a more conclusive comparison? Besides, a correlation of 0.36 does not seem so good?

We understand the Reviewer’s concern that the observed increase in the activity correlation may not appear to provide strong evidence for the improvement of the newly introduced model. However, within the context of polymer models developed to study realistic gene transcription and chromatin organization, this type of correlation analysis is a widely accepted approach for model validation. Experimental data commonly used for such validation include Hi-C maps, FISH experiments, and GRO-seq data [10,11]. The first two are typically employed to assess how accurately the model reproduces the 3D folding of chromatin; a comparison between experimental and simulated Hi-C maps is provided in the Supplementary Information (Fig. S5), showing a Pearson correlation of 0.7. GRO-seq or RNA-seq data, on the other hand, are used to evaluate the model’s ability to predict gene transcription levels. To date, the highest correlation for transcriptional activity data has been achieved by the HiP-HoP model at a resolution of 1 kbp [10], reporting a Spearman correlation of 0.6. Therefore, the correlation obtained with our 2-color model represents a good level of agreement when compared with the more complex HiP-HoP model. In this context, the observed increase in correlation—from 0.324 to 0.366—can be regarded as a modest yet meaningful improvement.

“...consequently, use of an additional color provides a statistically significant improvement (p -value $< 10^{-6}$, 2-sided t -test).” I do not follow this argument. Given enough simulation repeats, any improvement, no matter how small, will lead to statistically significant improvements.

We agree that this sentence could be misleading. We have now rephrased it in a clearer manner specifying that each of the two correlation values is statistically significant alone, while before we were wrongly referring to the significance of the improvement.

“Additionally, simulated contact maps show a fair agreement with Hi-C data (Figure S5), with a Pearson correlation $r \sim 0.7$ (p -value $< 10^{-6}$, 2-sided t -test).” Nice!

We thank the Reviewer for the positive comment.

“Because we do not include heterochromatin-binding proteins, we should not however expect a very accurate reproduction of Hi-C maps: we stress that here instead we are interested in active chromatin, transcription and structure only as far as it is linked to transcription.” Then why do you not limit your correlation assessment to only these regions to show that these are very well captured by your model?

We thank the Reviewer for this insightful comment. Indeed, we could have restricted our investigation to active chromatin regions, as done in our previous works [11,12]. However,

our intention in this section of the manuscript was to clarify that the current model is relatively simple and therefore not expected to achieve a very high level of agreement between experimental and simulated Hi-C maps. Another important limitation of the two color model described in the section is the absence of active loop extrusion mediated by SMC proteins, which is known to play a central role in establishing TADs boundaries. Consequently, even if our analysis were limited to active chromatin regions, the agreement with experimental Hi-C maps would still remain lower than that obtained with more comprehensive models, such as HiP-HoP, that we use later in the last section of the paper. We have now added a comment in the revised manuscript explicitly noting the lack of active loop extrusion in our 2-color model.

"We also measure the average value of the demixing coefficient, θ_{dem} (Materials and Methods). If $\theta_{dem} = 1$, this means that a cluster contains only TFs of one colour and so is fully demixed; if $\theta_{dem} = 0$, the cluster contains a mixture of TFs of all colors in equal number, and so is maximally mixed." Repetitive.

We have now rephrased the sentence in a more concise way.

"...notably, this is similar to the average number of productivelytranscribing pols seen experimentally in a transcription factory [6]." That seems a bit fast and loose. The number of Polymerases can differ depending on state, type of factory, gene etc. and vary between anything from to a few hundreds of Polymerase complexes depending on definition of factory, and what is counted as active. Also, one would think that polymerases only make up a small part of the overall protein pool that constitutes a condensate, so it is unclear whether this is a pertinent estimate.

Here we refer to the average size of what is normally referred to as a PolII factory, not a generic nuclear condensate. These are the clusters which arise in our simulations. These structures emerge through microphase separation and have been well characterised, for instance see [13] for a recent review. For these structures while there is a distribution the average is well defined and corresponds to a size of about 100 nm, which is very much in line with the size of the clusters we observe, both in terms of 3D diameter and number of participating proteins. Because of the size, the number of active complexes which can contribute cannot be significantly more than ~ 10 . These estimates are, we note, very much in line with super-resolution measurements of SAF-A clusters [14], which are associated with active transcription and hence it is reasonable to assume they colocalise with RNA and polymerase clusters.

"Conversely, activities of similar TUs lying far from each other on the genetic map are often weakly negatively correlated, as the formation of one cluster sequesters some TFs to reduce the number available to bind elsewhere." This point is interesting, and I strongly suspect that this indeed happening. But I don't think it was shown in the analysis of the simulation results in sufficient clarity. We need direct assessment of this sequestration, currently it's only indirectly inferred.

Indeed, this is the mechanism underlying the emergence of negative long-range correlations among TU activity values. As the Reviewer correctly pointed out, the competition for a finite number of TFs was only indirectly inferred in the original manuscript. To address this, we have now included a new figure explicitly illustrating this effect. In Fig. S12, we show the kymograph of active TUs (left panel), as in Fig. 2E(i) of the main text, alongside a new kymograph depicting the number of green TFs within a sphere of radius 10σ centered on each green TU (right panel). For simplicity, we focus here only on green TUs and TFs. It can be observed that, during the initial part of the simulation, green TFs are localized near genomic position ~ 2000 (right panel), where green TUs are transcriptionally active (left panel). Toward the end of the simulation, TUs near genomic position ~ 500 become active, coinciding with the relocation of TFs to this region and the depletion of the previous one.

In the definition for the demixing coefficient (equation 1), what does the index i stand for?

Here i is an index denoting each of the colors present in the model. We have now specified the meaning of i after Eq. 1.

Reviewer 3 (Public Review):

In this work, the authors present a chromatin polymer model with some specific pattern of transcription units (TUs) and diffusing TFs; they simulate the model and study TF clustering, mixing, gene expression activity, and their correlations. First, the authors designed a toy polymer with colored beads of a random type, placed periodically (every 30 beads, or 90kb). These colored beads are considered a transcription unit (TU). Same-colored TUs attract with each other mediated by similarly colored diffusing beads considered as TFs. This led to clustering (condensation of beads) and correlated (or anti-correlation) "gene expression" patterns. Beyond the toy model, when authors introduce TUs in a specific pattern, it leads to emergence of specialized and mixed cluster of different TFs. Human chromatin models with realistic distribution of TUs also lead to the mixing of TFs when cluster size is large.

Strengths.

This is a valuable polymer model for chromatin with a specific pattern of TUs and diffusing TF-like beads. Simulation of the model tests many interesting ideas. The simulation study is convincing and the results provide solid evidence showing the emergence of mixed and demixed TF clusters within the assumptions of the model.

Weaknesses.

Weakness of the work: The model has many assumptions. Some of the assumptions are a bit too simplistic. Concerns about the work are detailed below:

We thank the Referee for this overall positive evaluation.

We thank the Referee for this important observation. The way we The authors assume that when the diffusing beads (TFs) are near a TU, the gene expression starts. However, mammalian gene expression requires activation by enhancer-promoter looping and other related events. It is not a simple diffusion-limited event. Since many of the conclusions are derived from expression activity, will the results be affected by the lack of looping details?

We do not need to assume promoter-enhancer contact, this emerges naturally through the bridging-induced phase separation and indeed is a key strength of our model. Even though looping is not assumed as key to transcriptional initiation, in practice the vast majority of events in which a TF is near a TU are associated with the presence of a cluster where regulatory elements are looped. So transcription in our case is associated with the bridging-induced phase separation, and there is no lack of looping, looping is naturally associated with transcription, and this is an emergent property of the model (not an assumption), which is an important feature of our model. Accordingly, both contact maps and transcriptional activity are well predicted by our model, both in the version described here and in the more sophisticated single-colour HiP-HoP model [10] (an important ingredient of which is the bridging-induced phase separation).

Authors neglect protein-protein interactions. Without protein-protein interactions, condensate formation in natural systems is unlikely to happen.

We thank the Reviewer for pointing out the absence of protein-protein interactions in our simulations. While we acknowledge this limitation, we would like to emphasize that experimental studies have not observed nuclear proteins forming condensates at physiological concentrations in the absence of DNA or chromatin. For example, studies such as Ryu et al. [15] and Shakya et al. [16] show that protein-protein interactions alone are insufficient to drive condensate formation in vivo. Instead, the presence of a substrate, such as DNA or chromatin, is essential to favor and stabilize the formation of protein clusters.

In our simulations, we propose that protein liquid-liquid phase separation (LLPS) is driven by the presence of both strong and weak attractions between multivalent protein complexes and the chromatin filament. As stated in our manuscript, the mechanism leading to protein cluster formation is the bridging induced attraction. This mechanism involves a positive feedback loop, where protein binding to chromatin induces a local increase in chromatin density, which then attracts more proteins, further promoting cluster formation.

While we acknowledge that adding protein-protein interactions could be incorporated into our simulations, we believe this would need to be a weak interaction to remain consistent with experimental data. Additionally, incorporating such interactions would not alter the conclusions of our study.

What is described in this paper is a generic phenomenon; many kinds of multivalent chromatin-binding proteins can form condensates/clusters as described here. For example, if we replace different color TUs with different histone modifications and different TFs with Hp1, PRC1/2, etc, the results would remain the same, wouldn't they? What is specific about transcription factor or transcription here in this model? What is the logic of considering 3kb chromatin as having a size of 30 nm? See Kadam et al. (Nature Communications 2023). Also, DNA paint experimental measurement of 5kb chromatin is greater than 100 nm (see work by Boettiger et al.).

We thank the Reviewer for this important observation, which we now address. To begin, we consider the toy model introduced in the first part of the manuscript, where TUs are randomly positioned rather than derived from epigenetic data. As the Reviewer points out, in this simplified context, our results reflect a generic phenomenon: the composition of clusters depends primarily on their size, independent of the specific types of proteins involved. However, the main goal of our work is to gain insights into apparently contradictory experimental findings, which show that some transcription factories consist of a single type of transcription factors, while other contain multiple types. This led us to focus on TF clusters and their role in transcriptional regulation and co-regulation of distant genes. Therefore, in the second part of the manuscript, we use DNase I hypersensitive site (DHS) data to position TUs based on predicted TF binding sites, providing a more biological framework. In both the toy model and the more realistic HiP-HoP model, we observe a size-dependent transition in cluster composition. However, we refrain from generalizing these results to clusters composed of other protein complexes, such as HP1 and PRC, as their binding is governed by distinct epigenetic marks (e.g. *H3K927me3* and *H3K27me3*), which exhibit different genomic distributions compared to DHS marks.

Finally, the mapping of 3kb to 30nm is an estimate which does not significantly impact our conclusions. The relationship between genomic distance (in kbp) and spatial distance (in nm) is highly dependent on the degree of chromatin compaction, which can vary across cell types and genomic context. As such, providing an exact conversion is challenging [17]. For example, in a previous work based on the HiP-HoP model [12] we compared simulated and experimental FISH measurements and found that 1kbp typically corresponds to 15 – 20nm, implying that 3kbp could span 60nm. Nevertheless, we emphasize that varying this conversion factor does not affect the core results or conclusions of our study. We have now included a clarification in the revised SI to highlight this point.

Recommendations for the authors:

Other points.

Figure 1(D) caption says $2.25\sigma = 1.6$ nanometer. Is this a typo? Sigma is 30nm.

Yes, it was. As $1\sigma \sim 30\text{nm}$, we have $2.25\sigma = 2.25 \cdot 30 \text{ nm} = 67.2 \text{ nm} \sim 6.7 \times 10^{-8}\text{m}$. We have now corrected the caption.

Page 6, column 2nd, 3rd para, it is written that θ_{dem} ("defined in Fig.1"). There is no θ_{dem} defined in Fig.1, is there? I can see it defined in Methods but not in Fig. 1.

Correct, we replaced (defined in Fig.1) with (see Methods for definition).

Page 6, column 2, 4th para: what does "correlations overlap and correlations diverge mean"?

With reference to the plots from Fig. 5B, correlation overlap and diverge simply refers to the fact that same-colour (red curves) and different-colour (blue curves) correlation trends may or may not overlap on each other. We have now clarified this point.

What is the precise definition of correlation in Fig 5B (Y-axis)?

In Fig.5B, correlation means Pearson correlation. We have now specified this point in the revised text and in the caption of Fig.5.

References

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